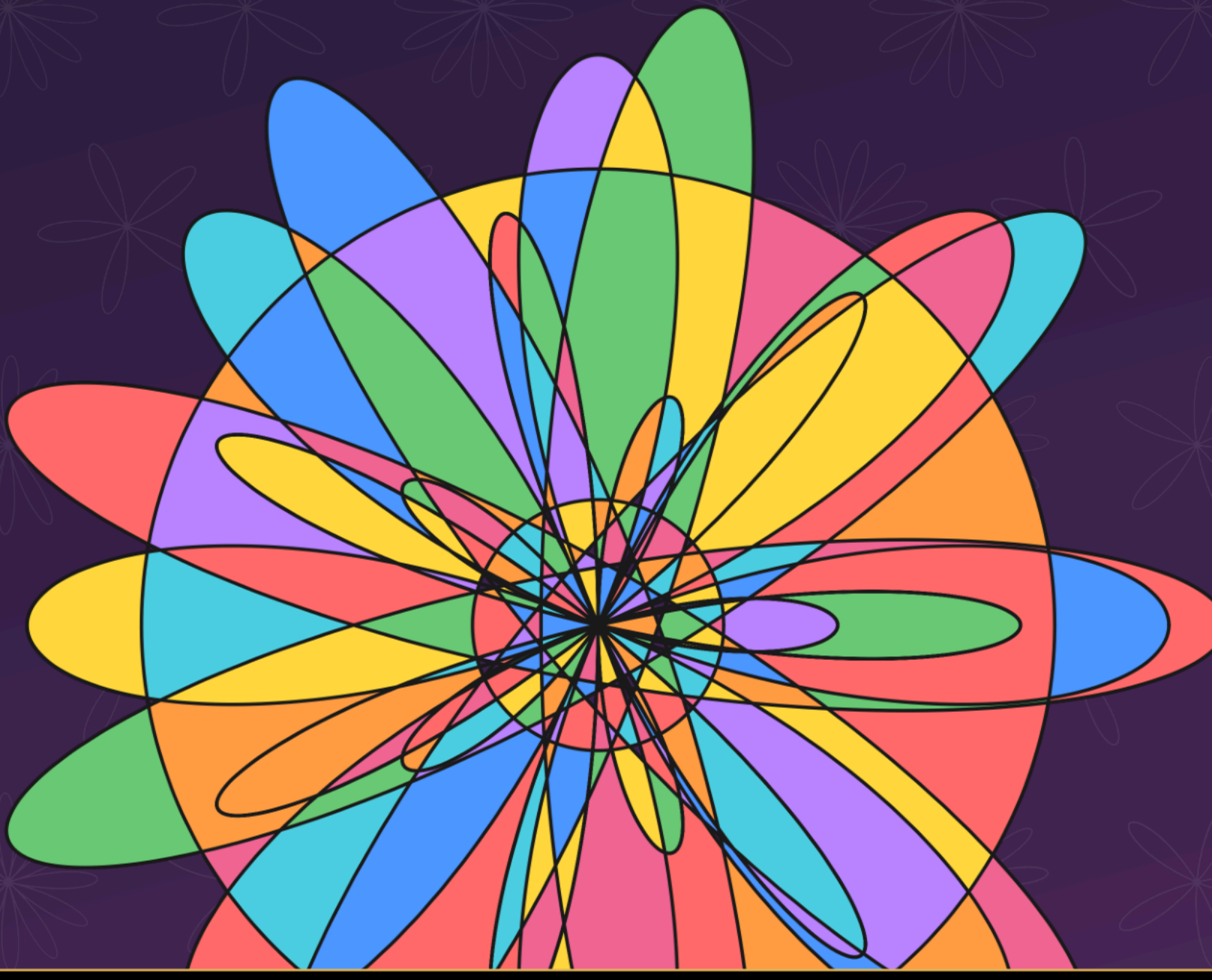
MATHEMATICAL TANGLE ART

A Coloring Book of Curves



MathRelaxing

Mathematical Tangle Art

A Coloring Book of Curves

mathrelaxing

(c) 2026 mathrelaxing

mathrelaxing.store

All rights reserved. No part of this book may be reproduced without written permission from the publisher.

First edition

How to Colour This Book

Your paper: 60# (90 gsm) uncoated white — a smooth, lightly textured matte sheet that loves dry media.

Best choice — Coloured pencils

Wax- or oil-based pencils are ideal on this paper. Build colour from light to dark in thin layers, then burnish with a colourless blender or a pale pencil for a smooth, even finish.

Excellent — Fineliners & gel pens

Use fine tips (0.3–0.5 mm) for the smallest regions. Let the ink settle for a moment before resting your hand so it doesn't smudge. Metallic and white gel pens shine for highlights.

Use with care — Markers

Alcohol- and water-based markers will bleed through a 90 gsm sheet. Slip a spare sheet of paper behind the page you're working on and colour one side at a time.

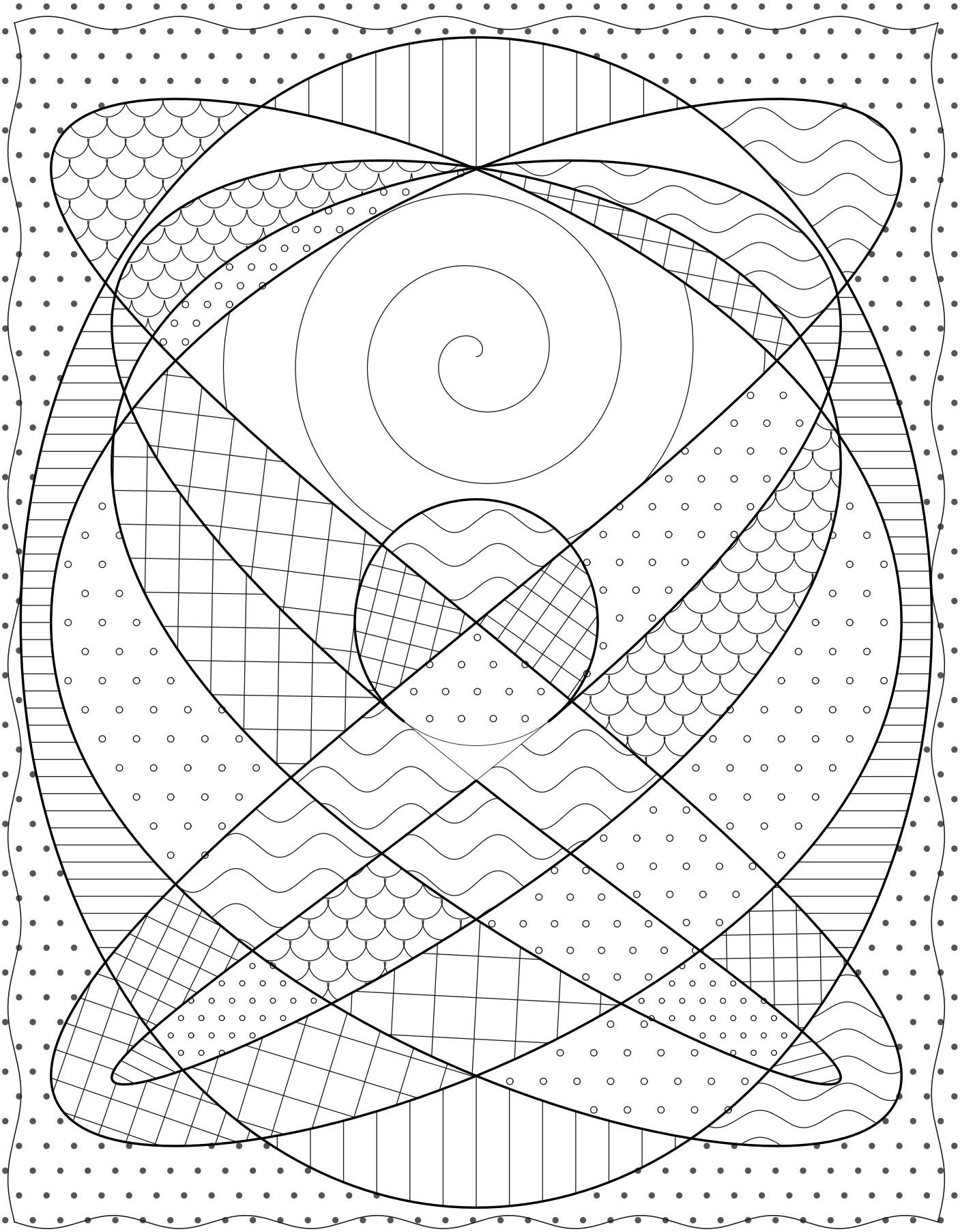
Better to avoid — Wet media

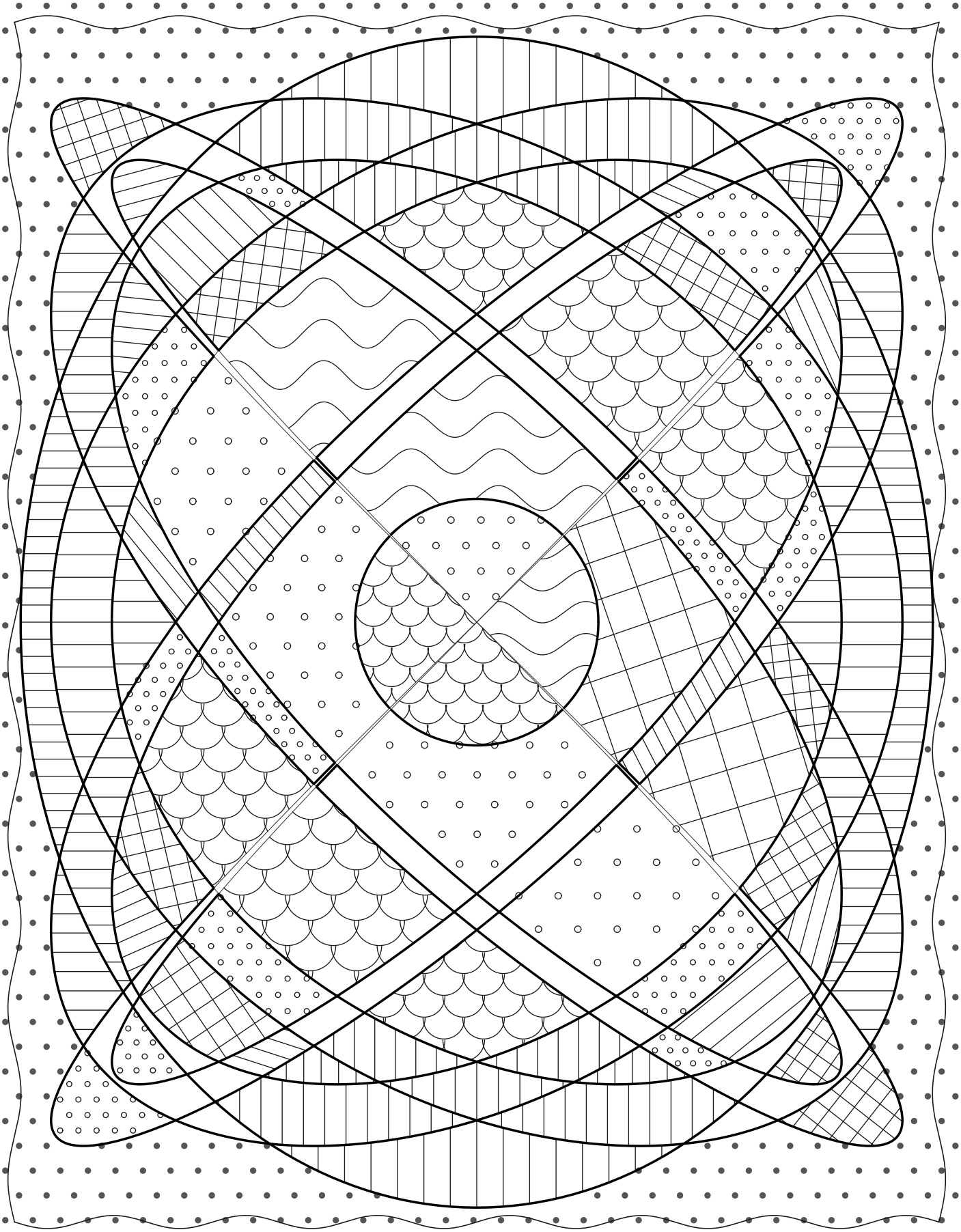
Watercolour, gouache and heavy washes soak and buckle thin paper. If you love paint, photocopy the design onto thick card stock first.

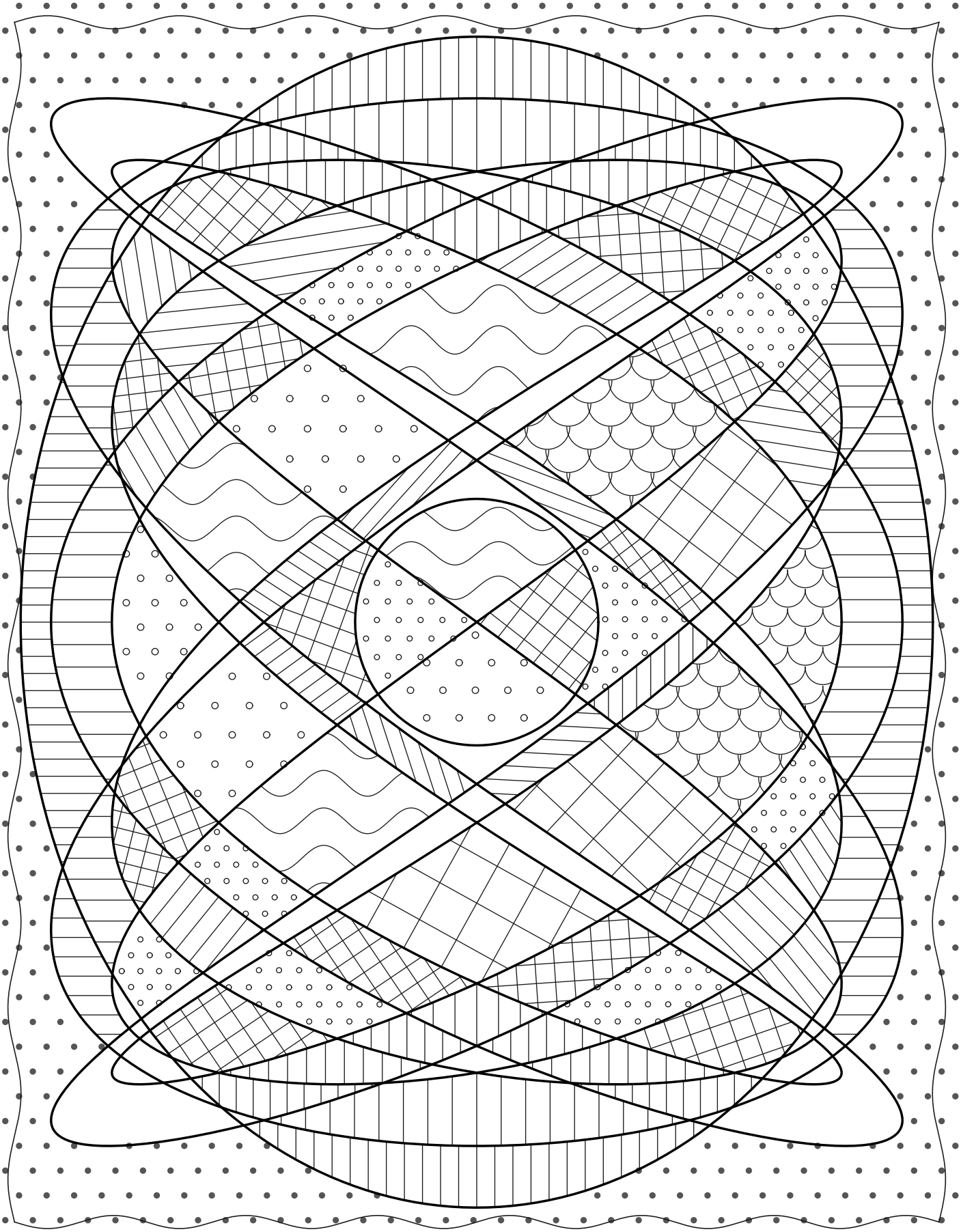
A few gentle reminders

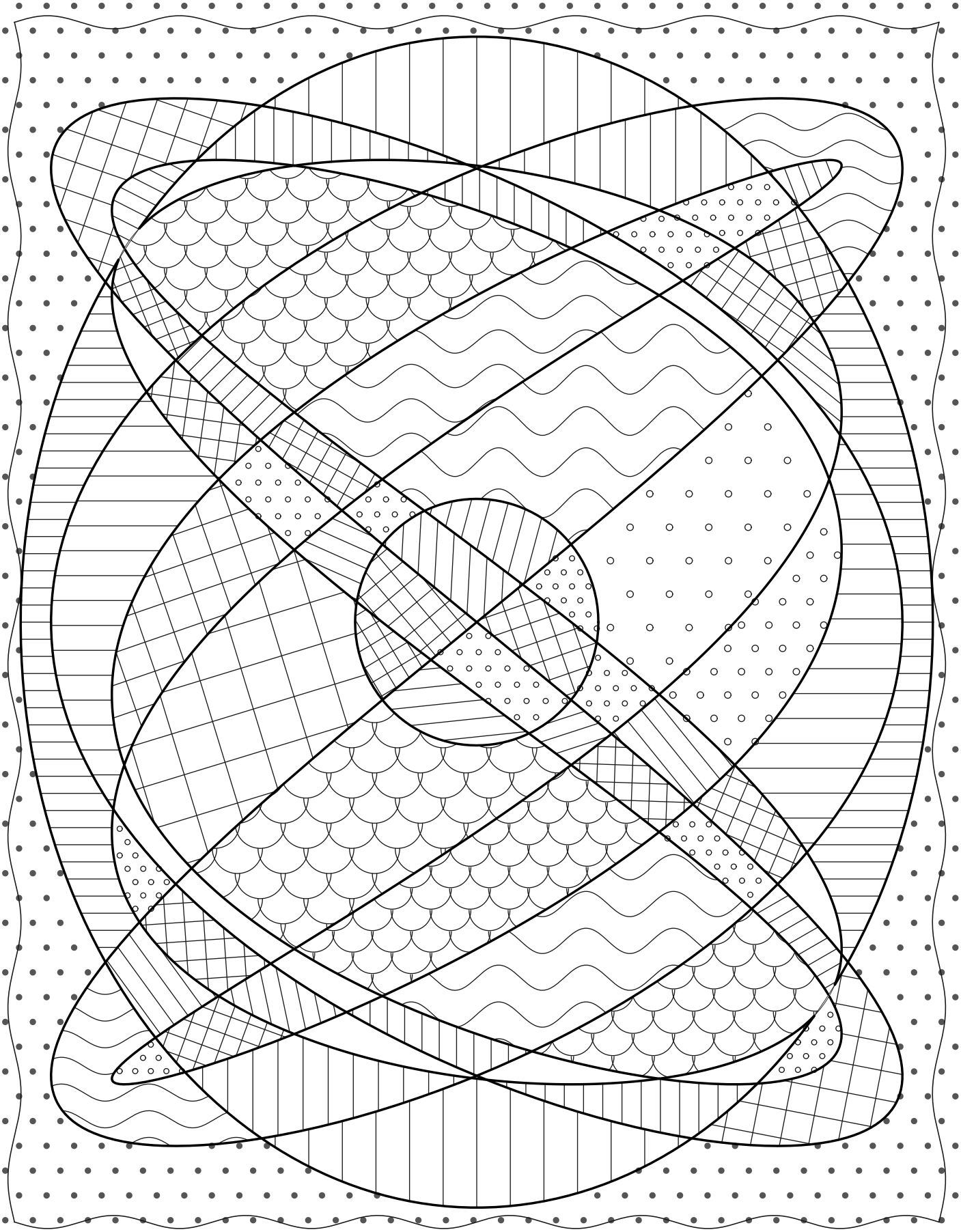
- Rest your hand on a scrap sheet to keep the page clean.
- Colour in good, natural light to judge your tones.
- Start at the centre and work outward to avoid smudging.
- There is no wrong way — the point is to slow down and relax.

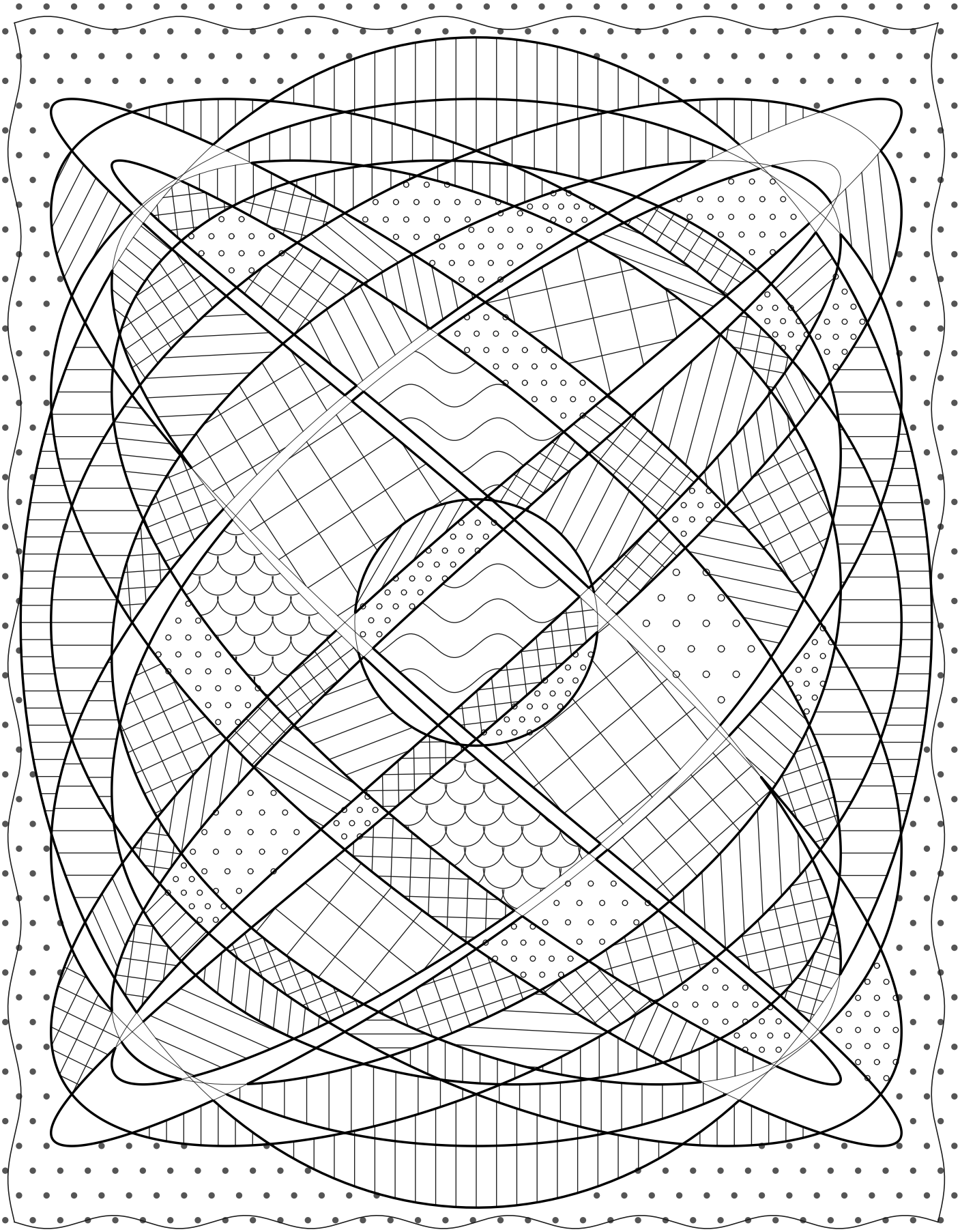


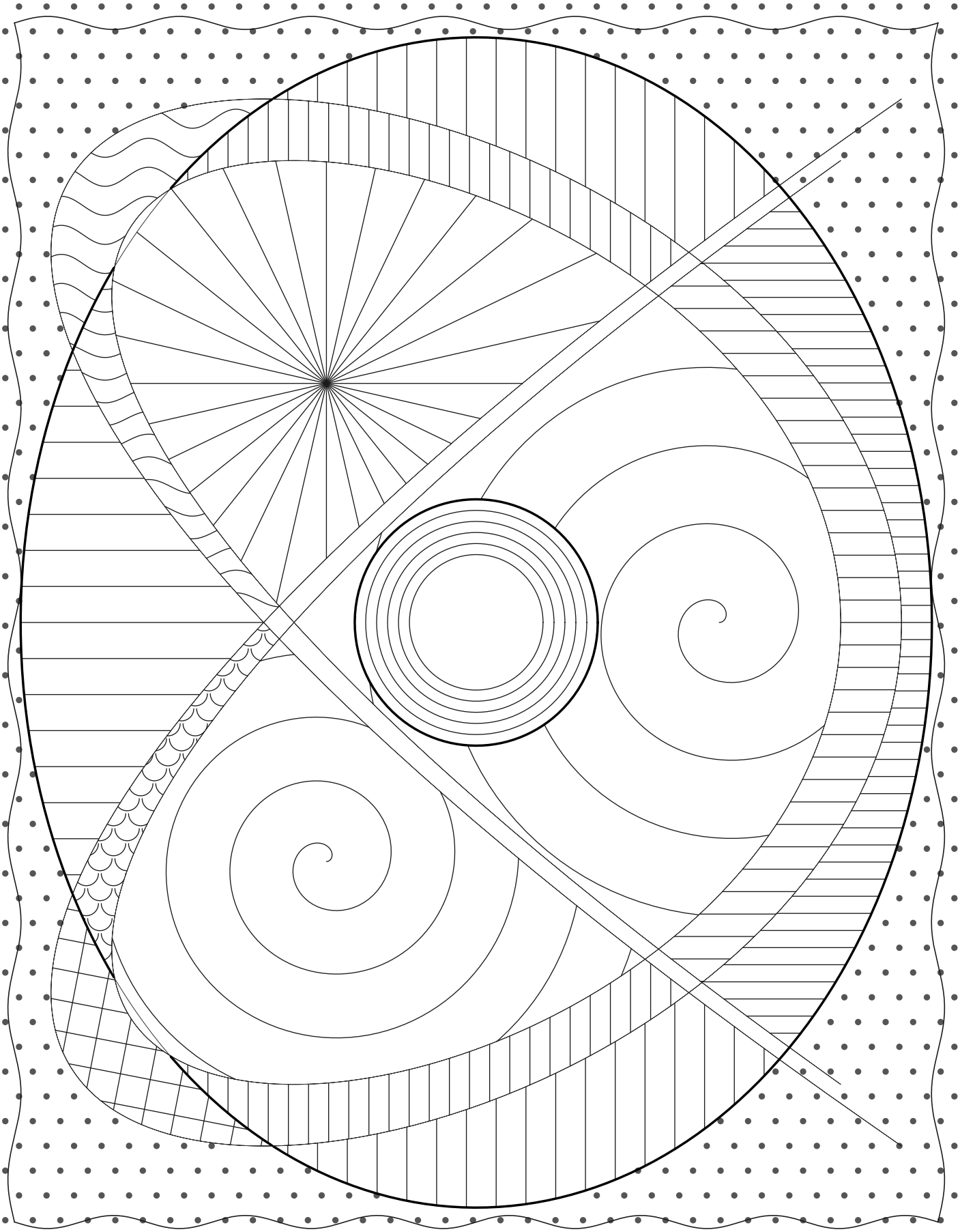


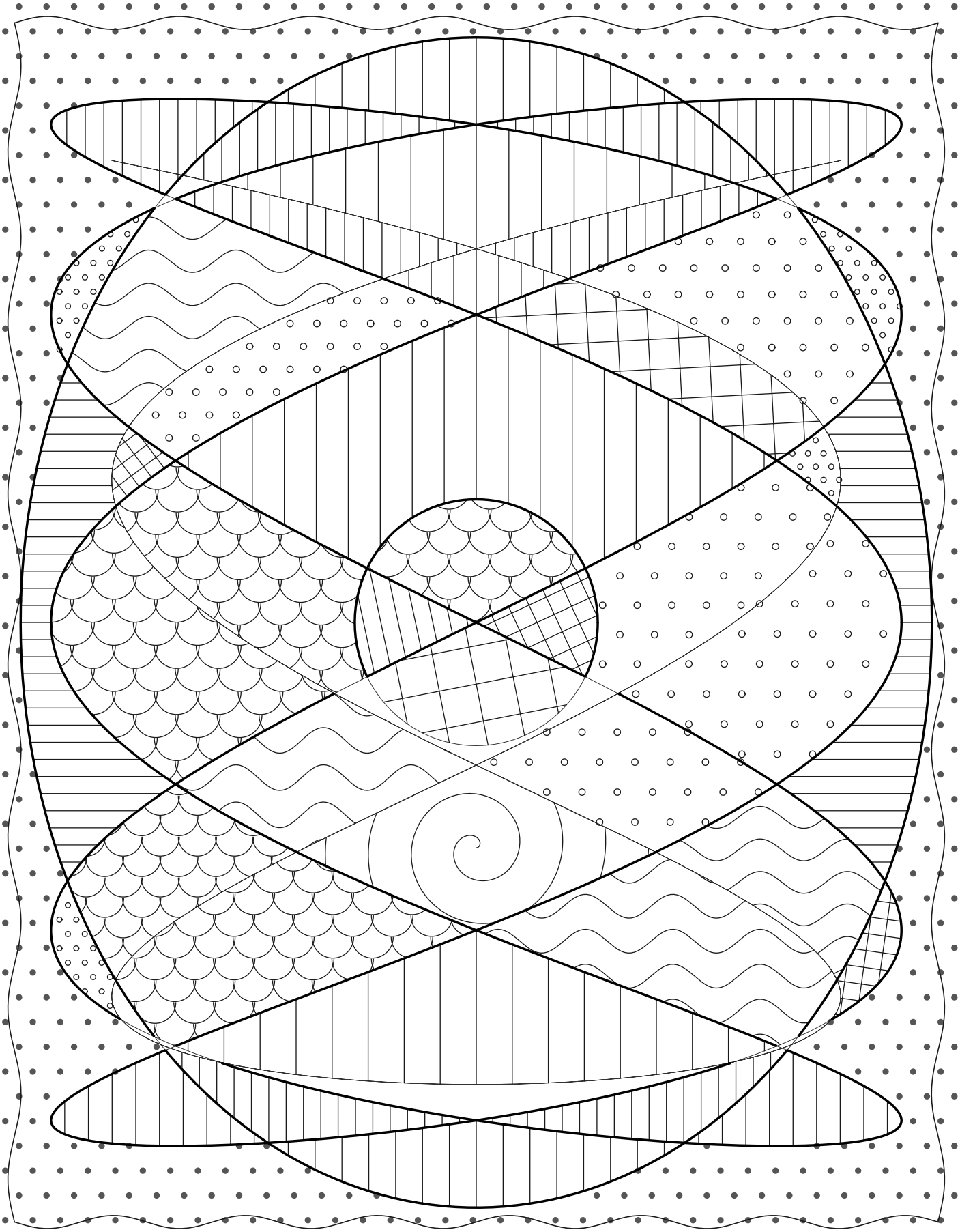


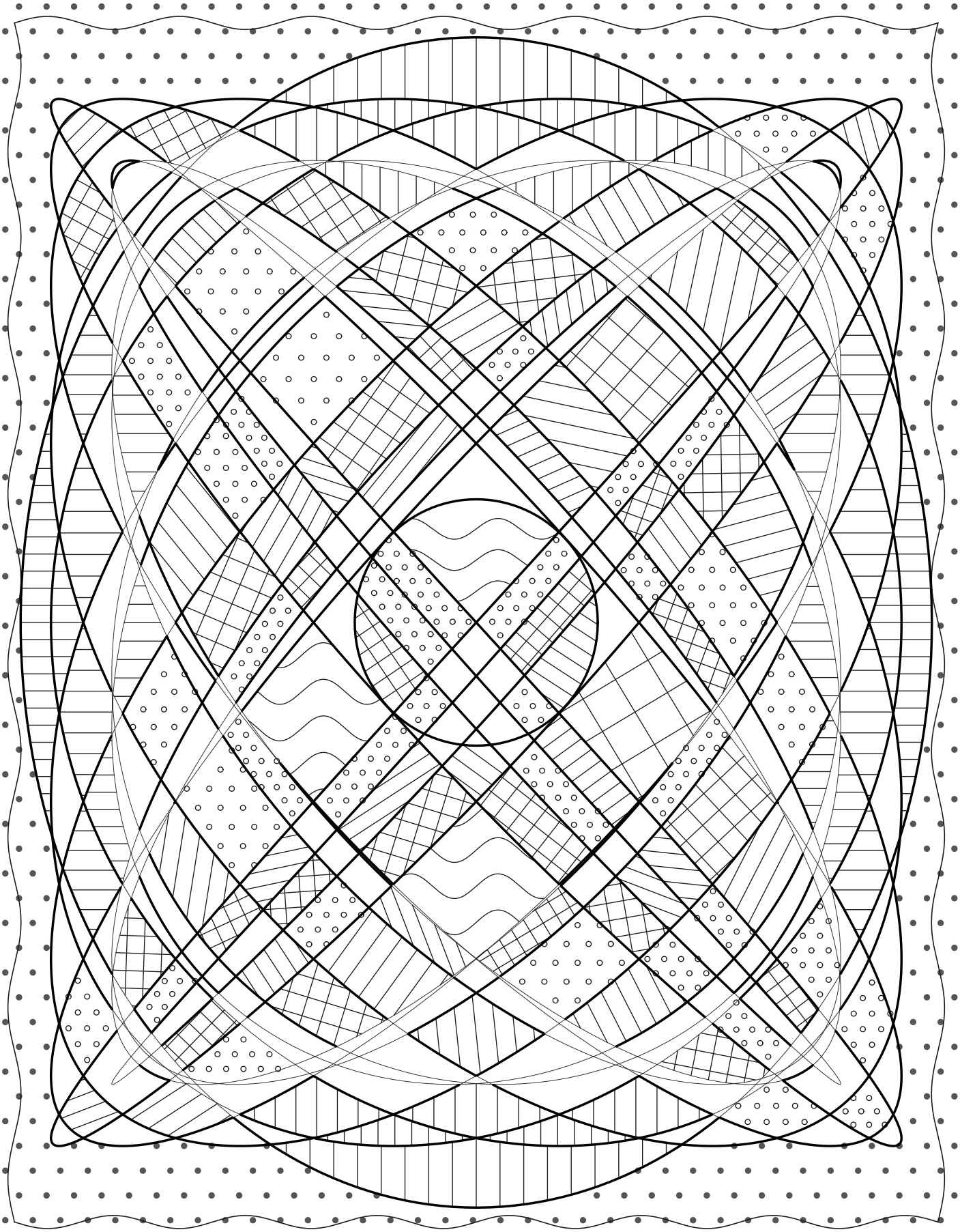


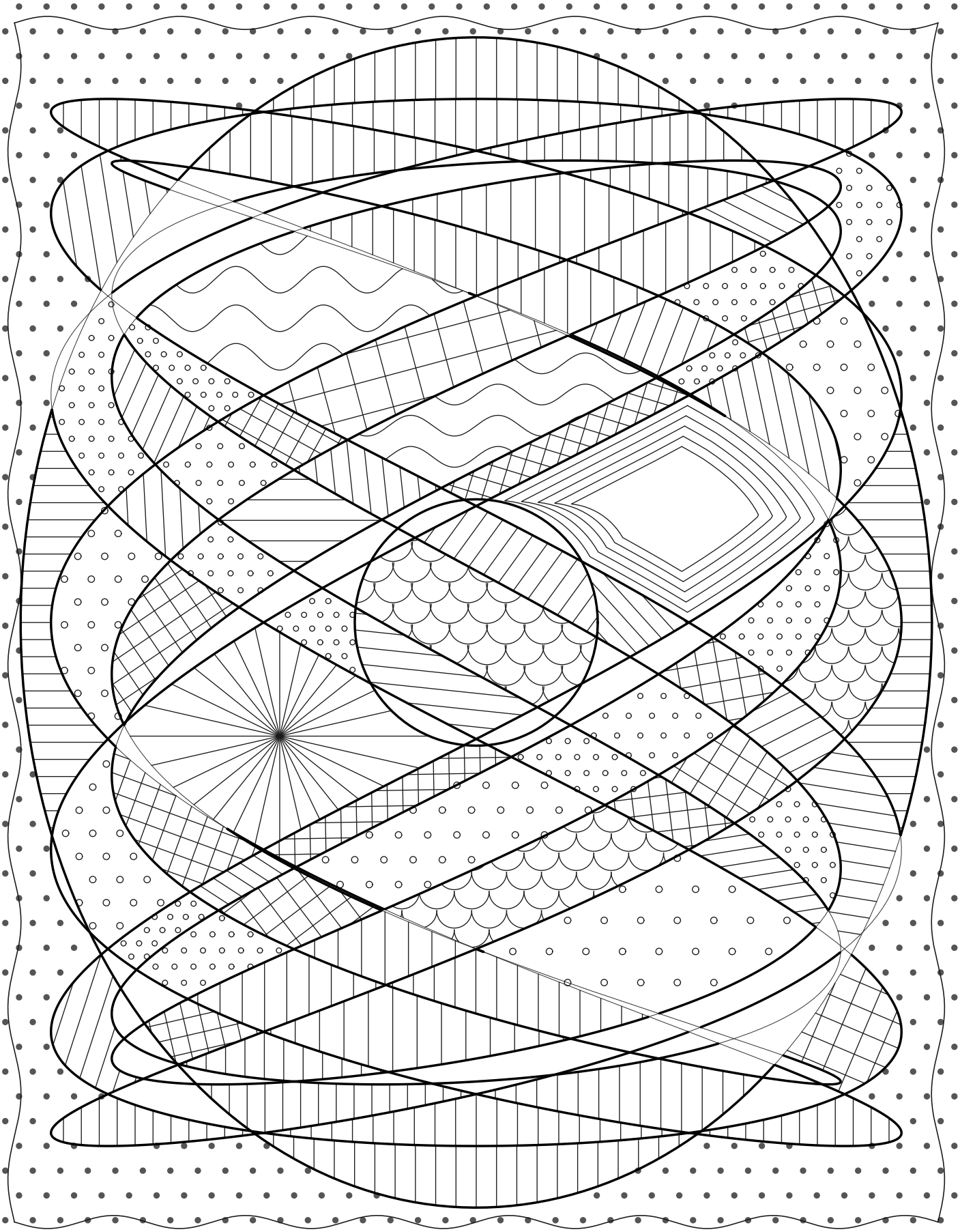


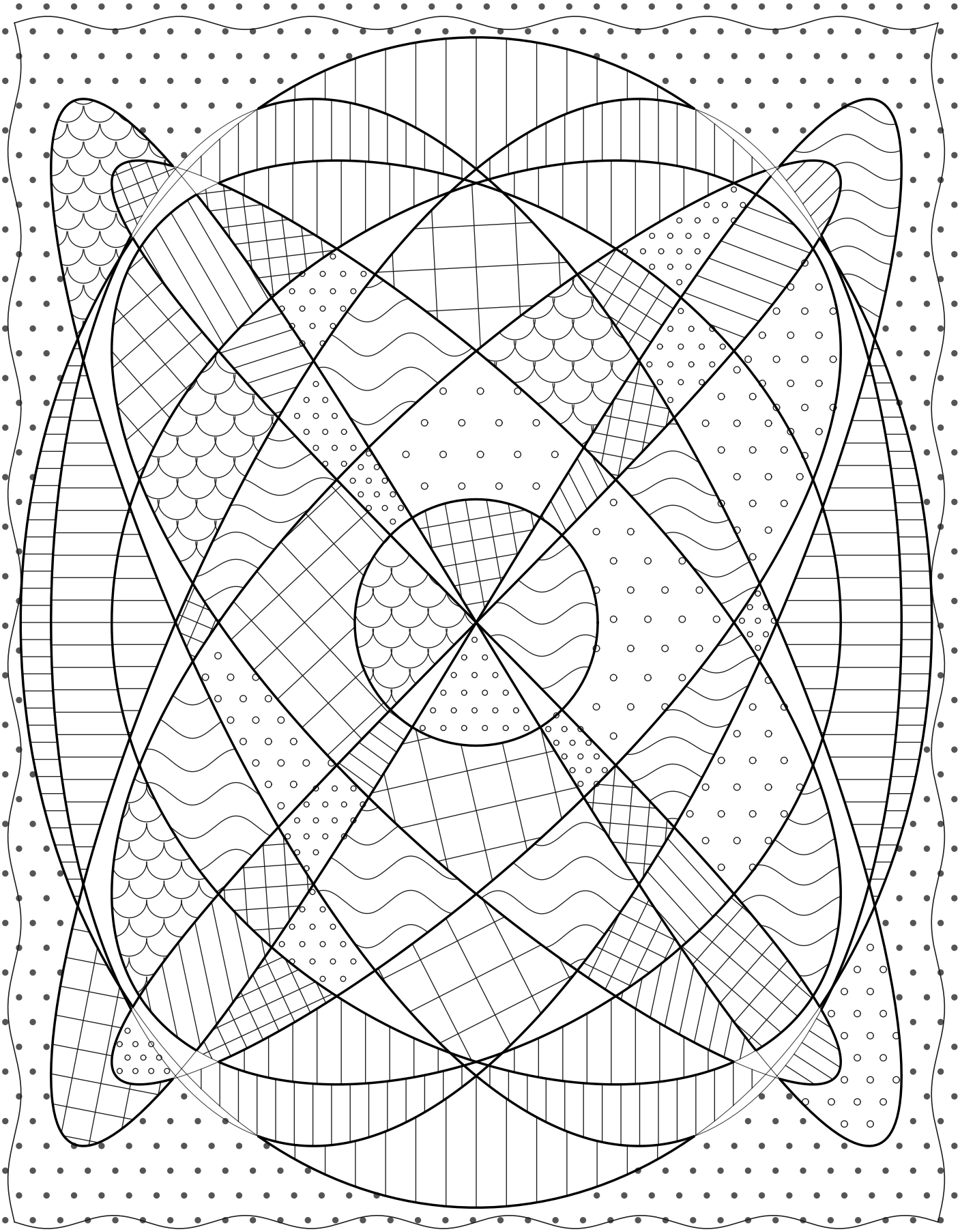


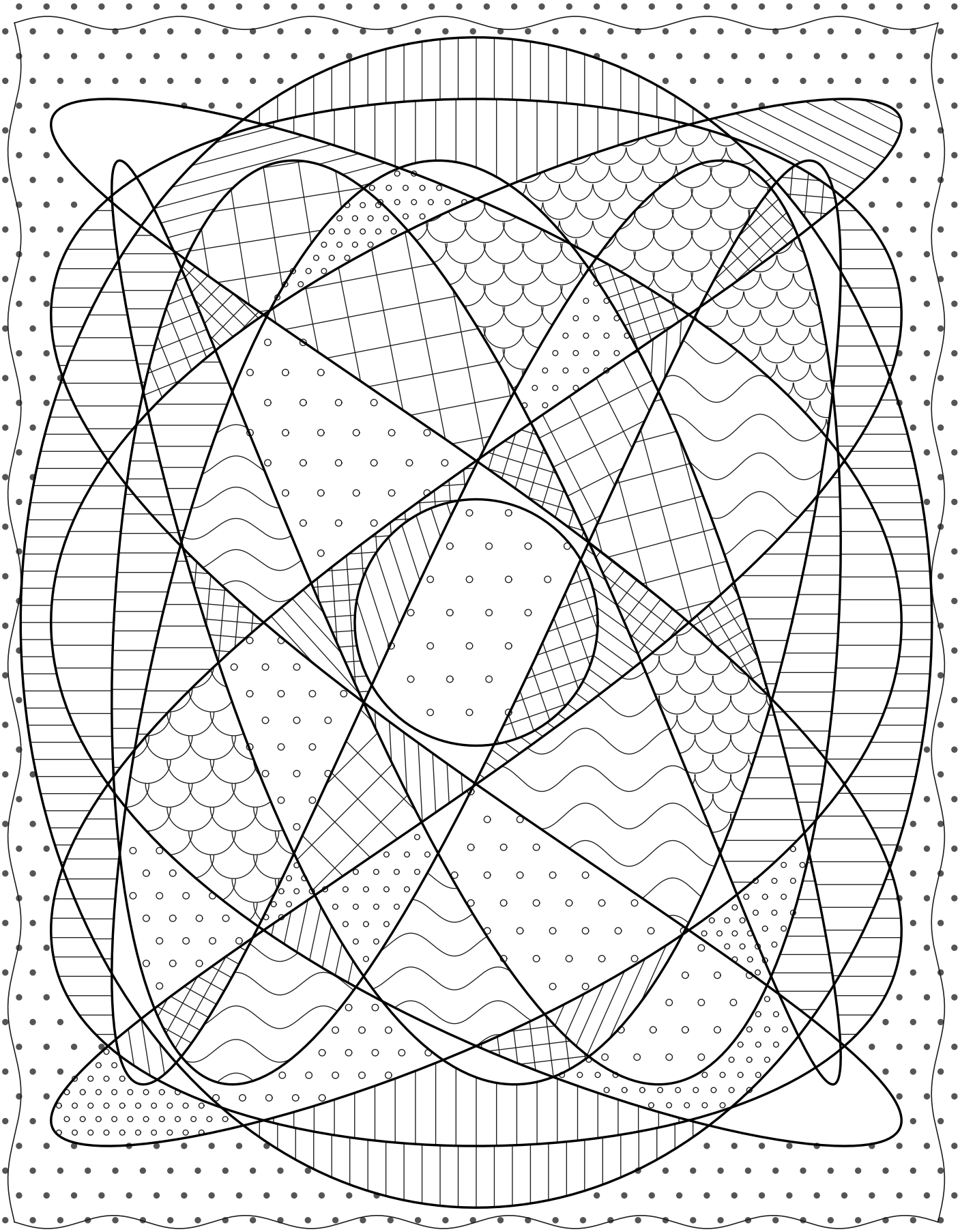


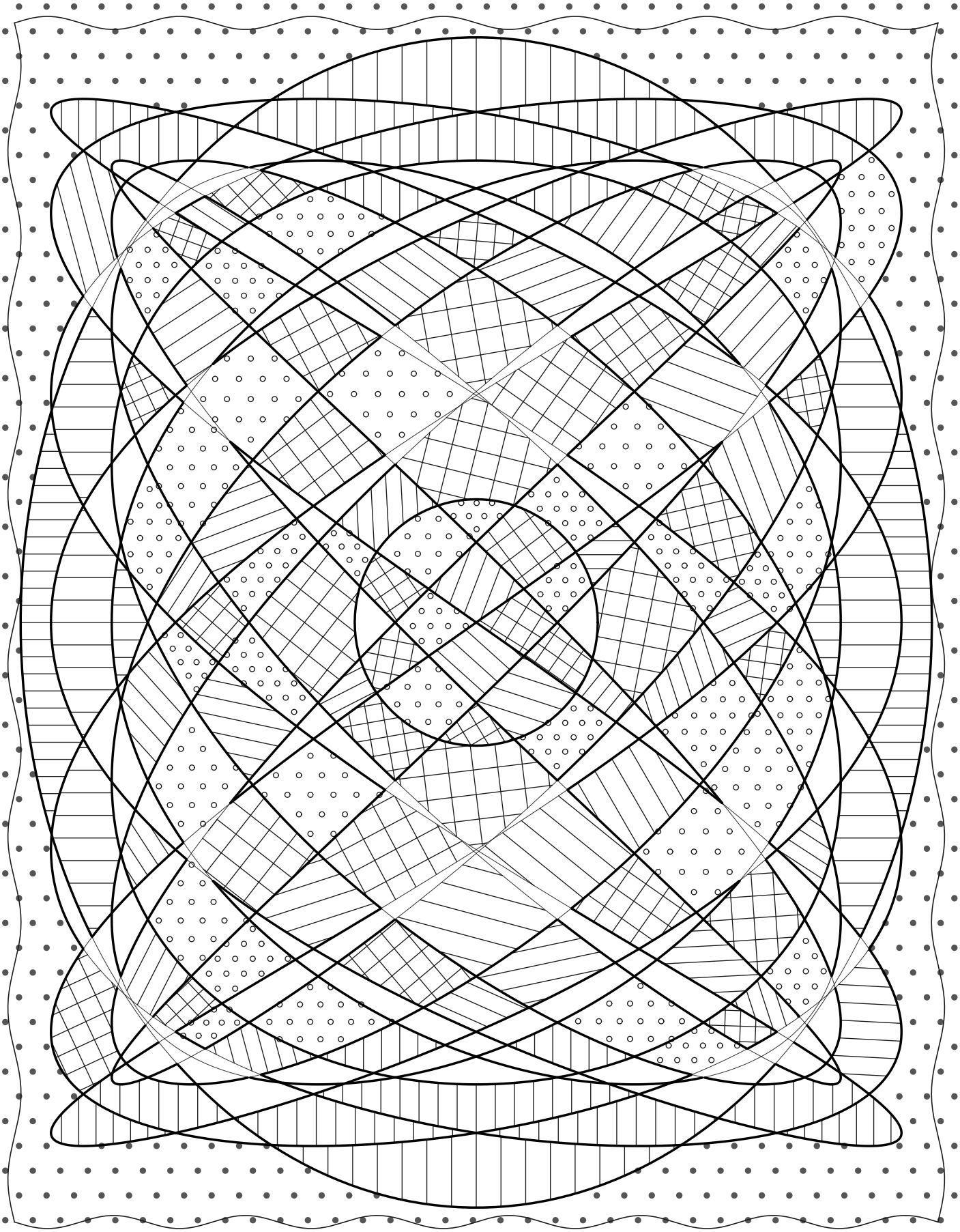


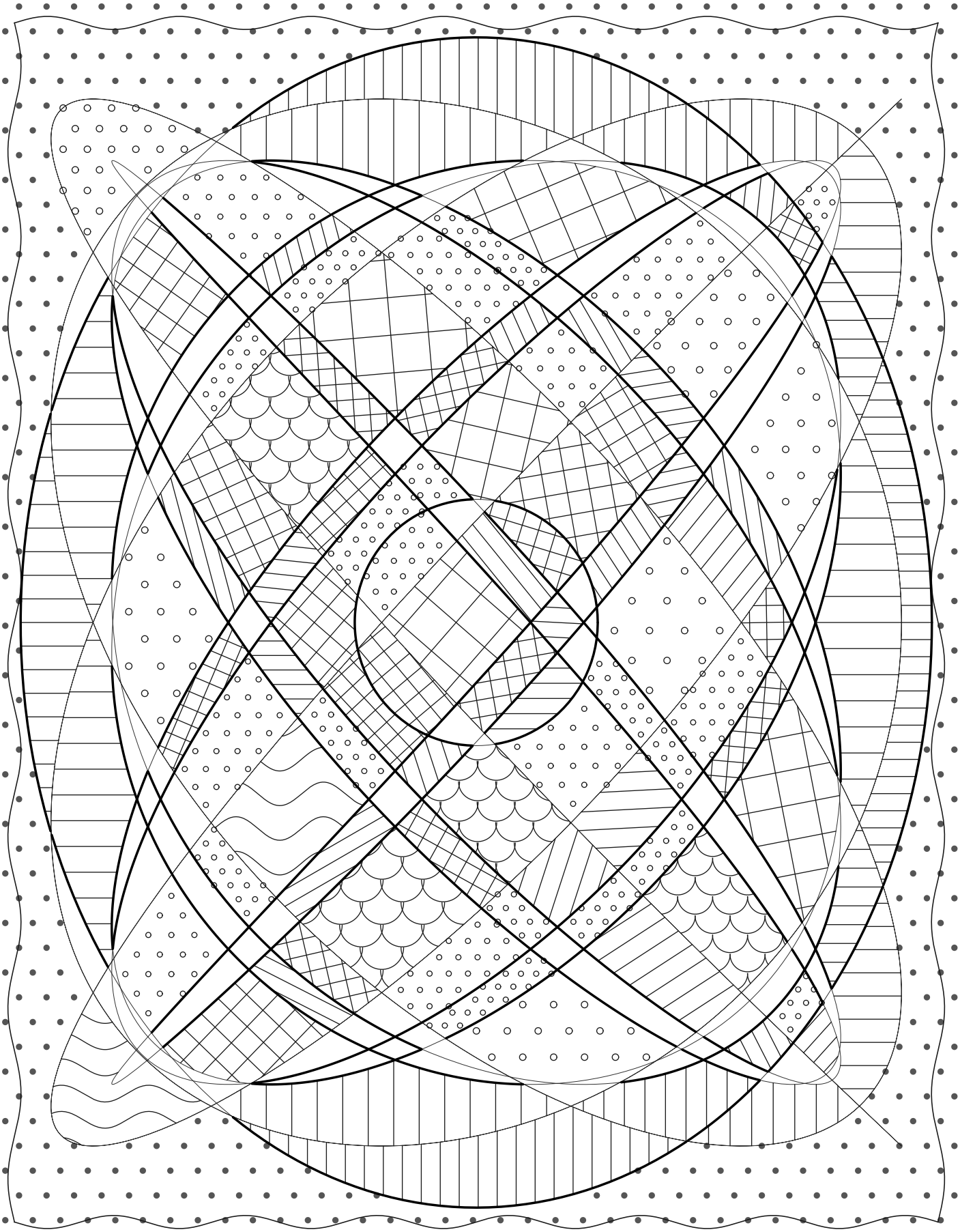


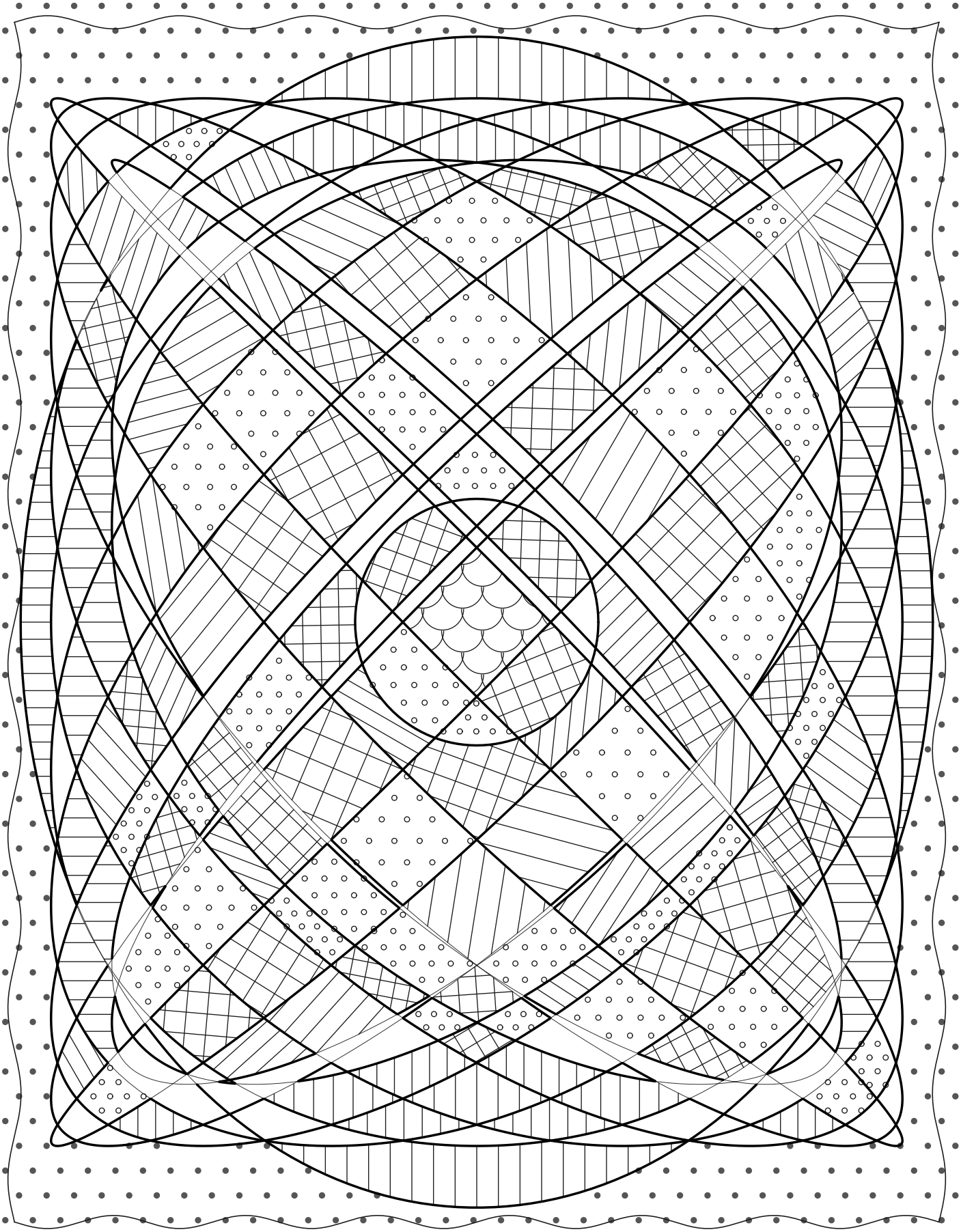


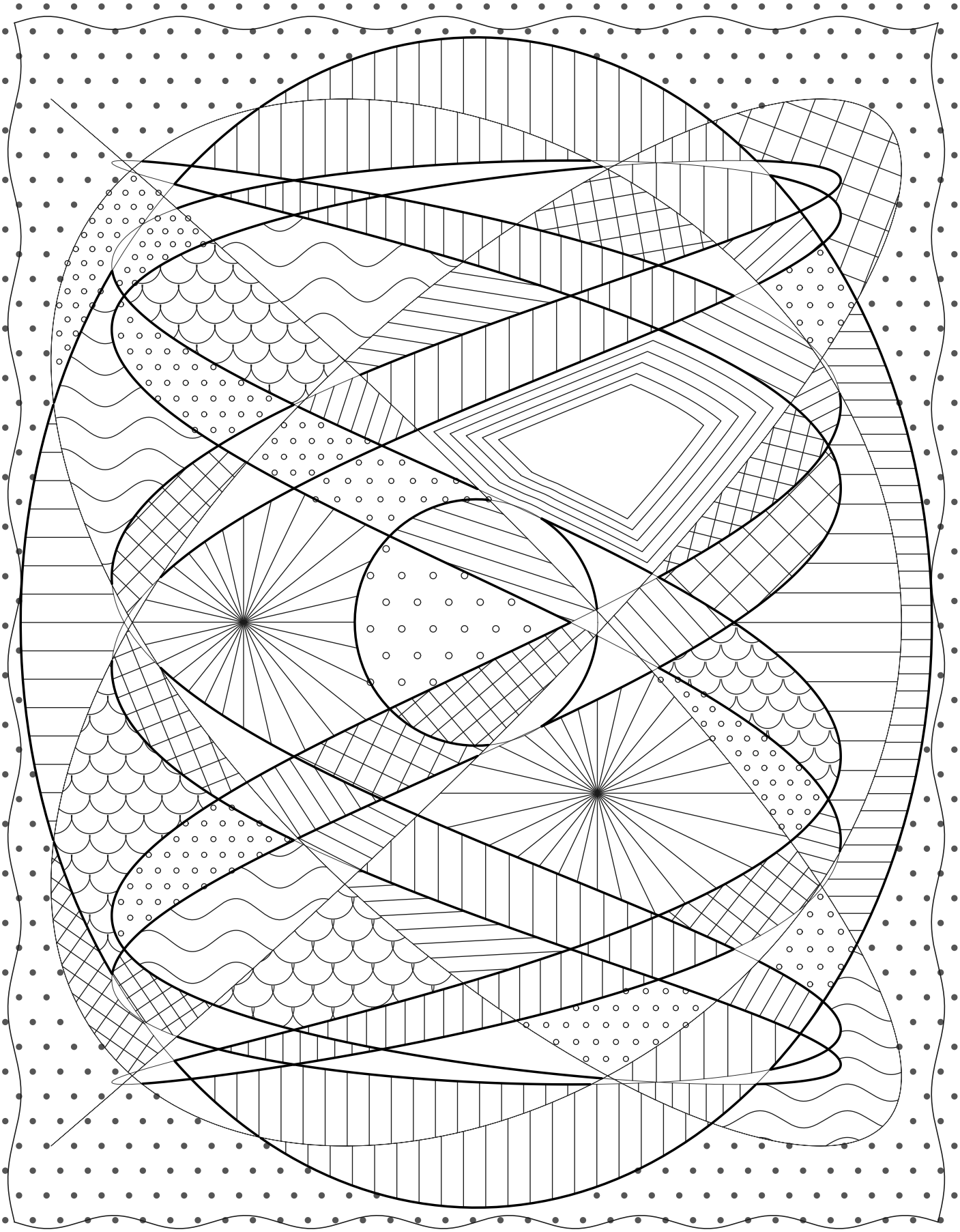


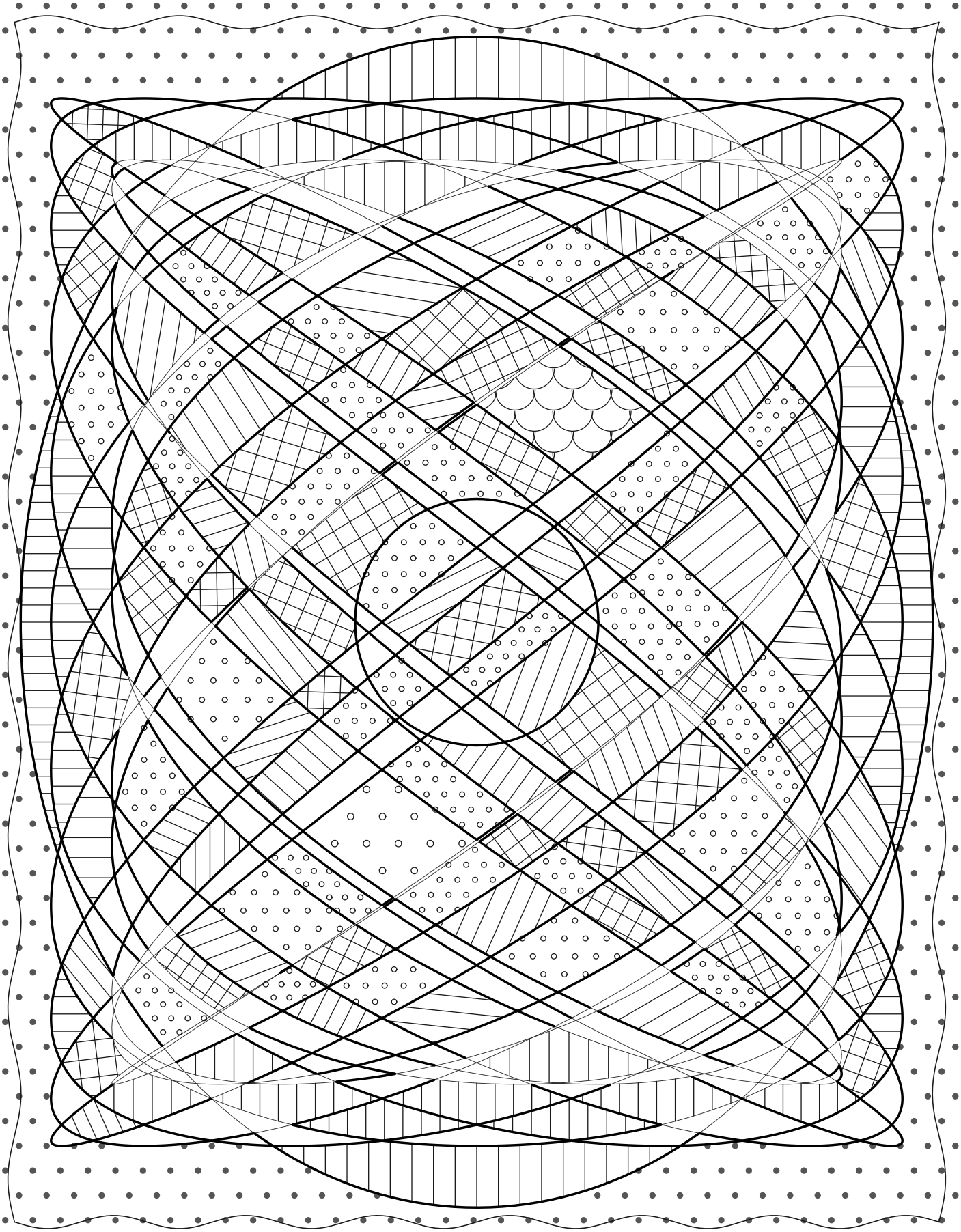


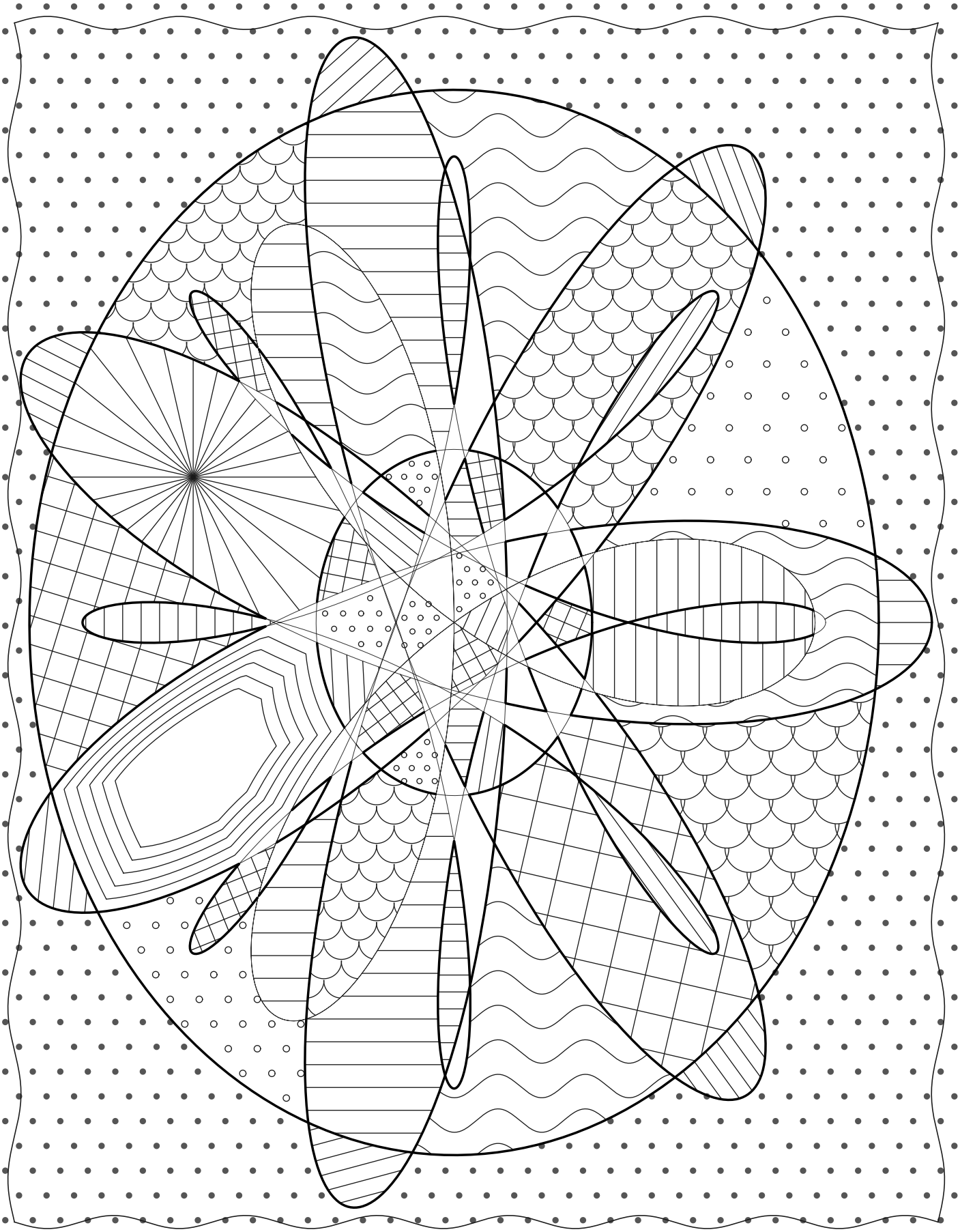


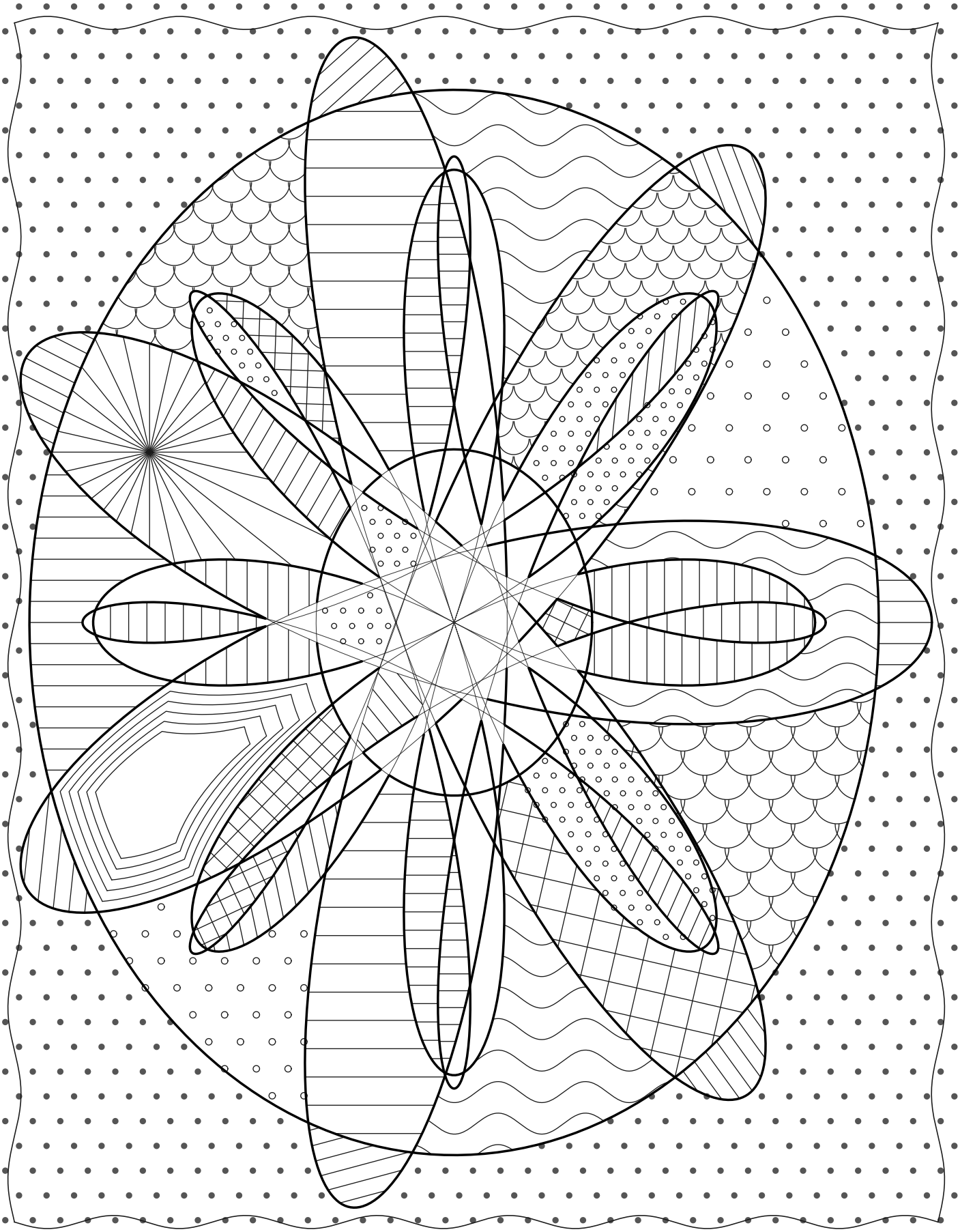


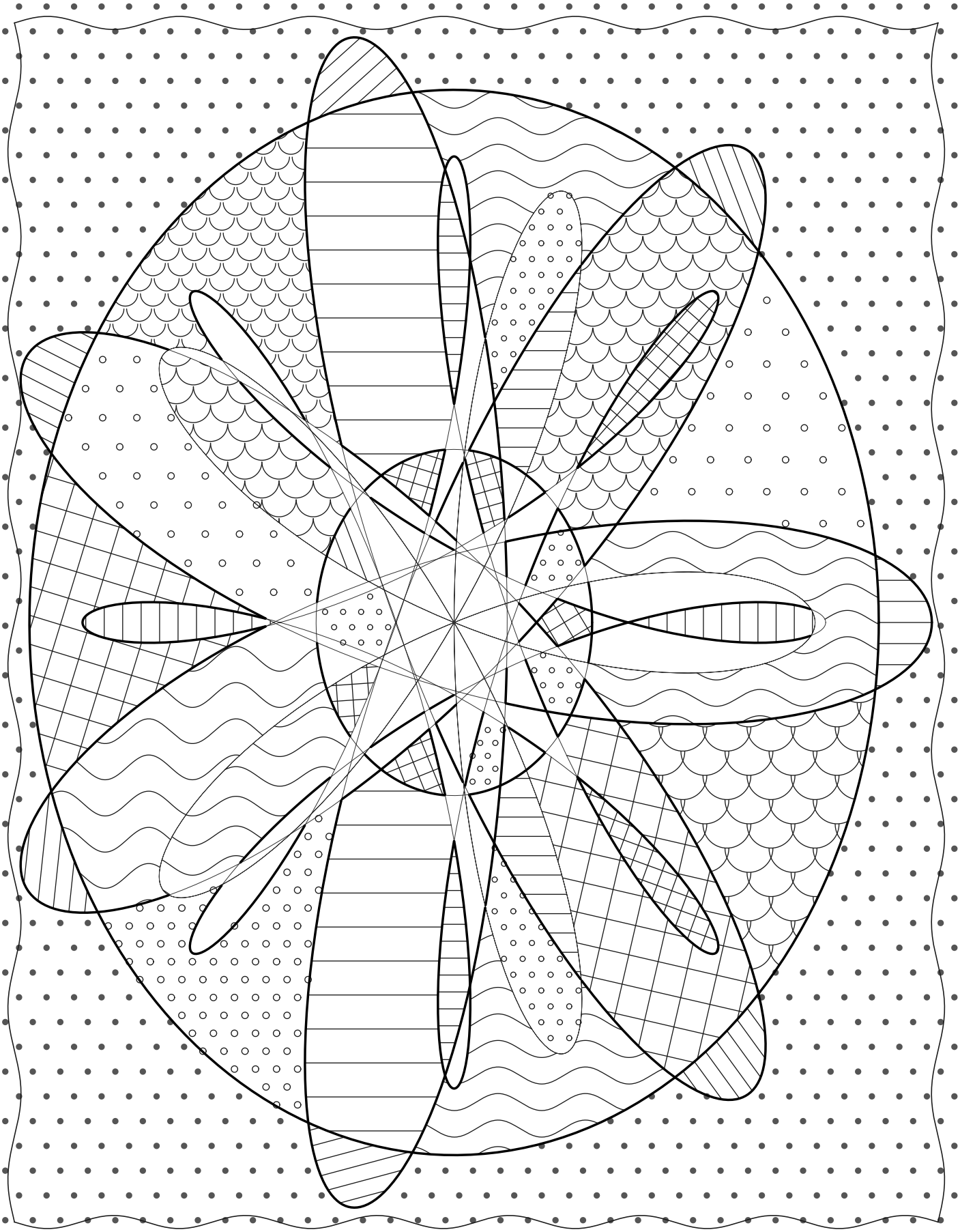


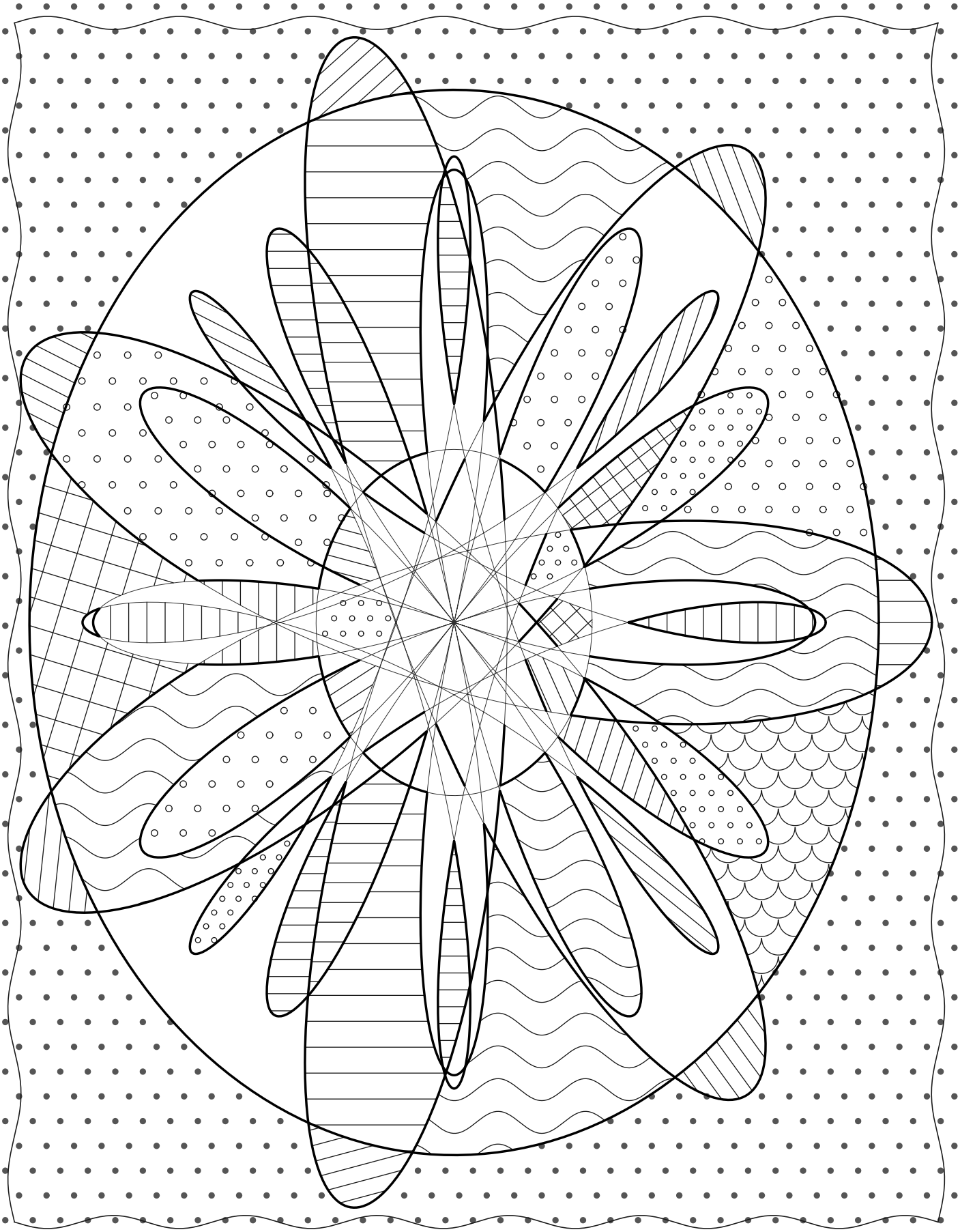


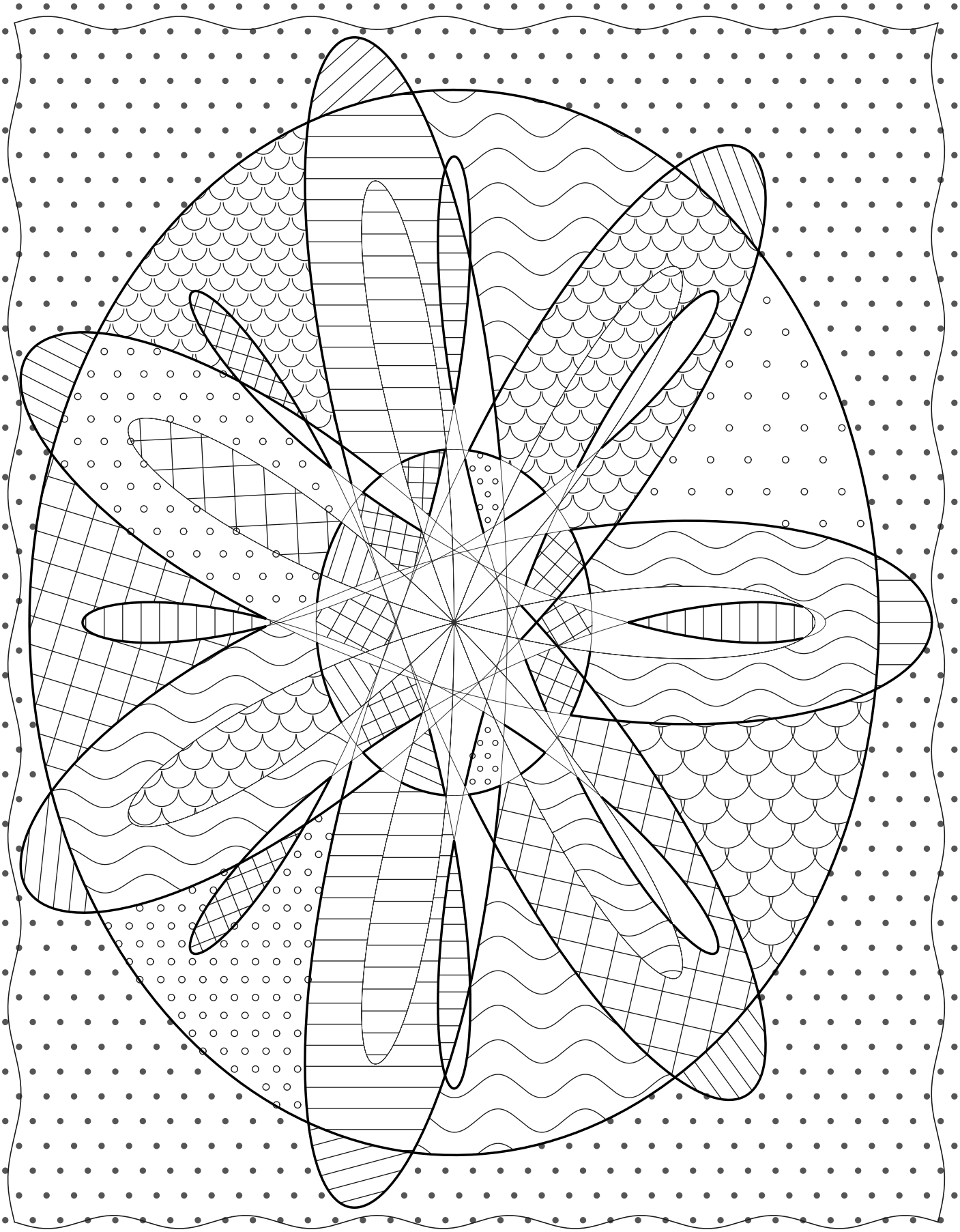


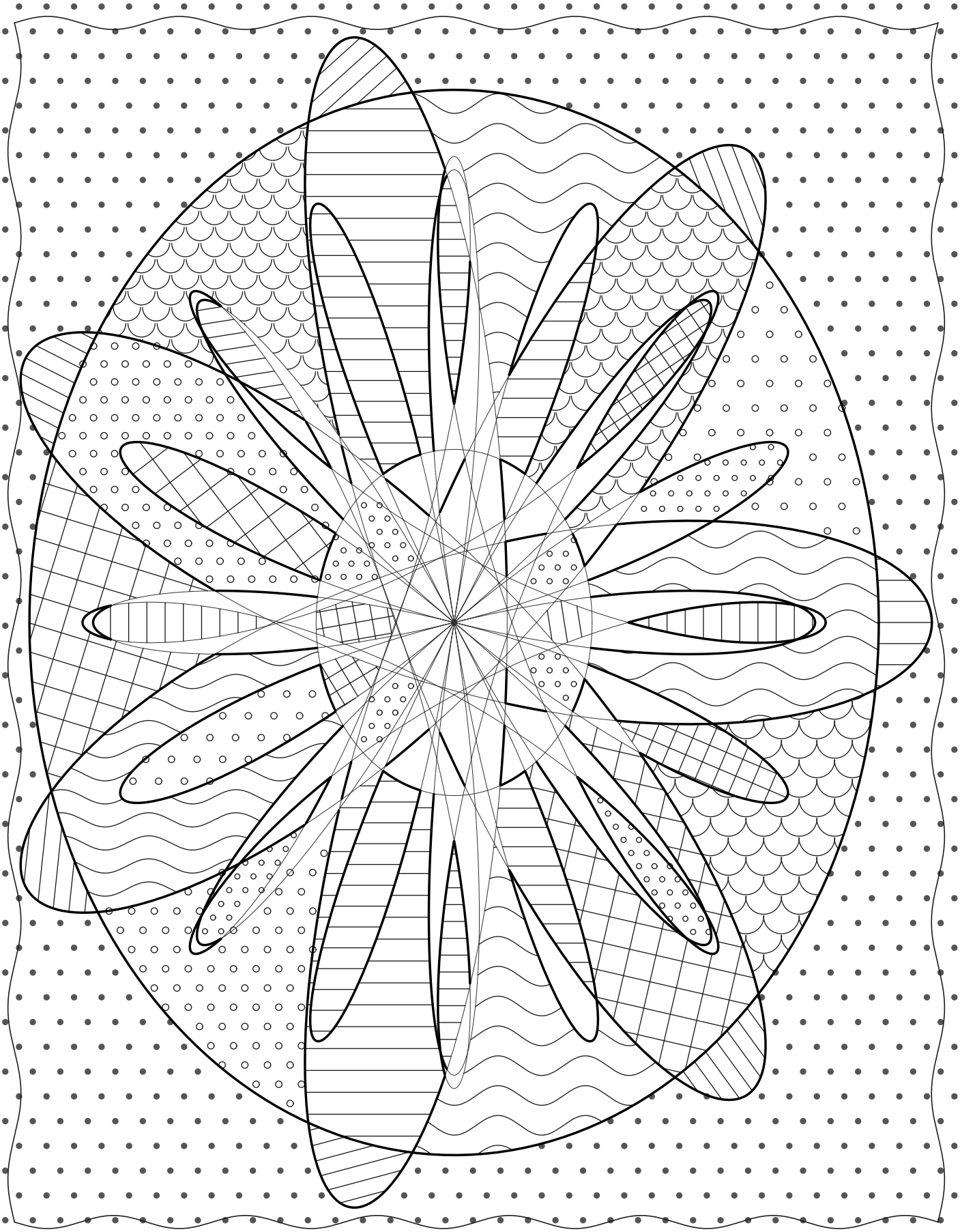


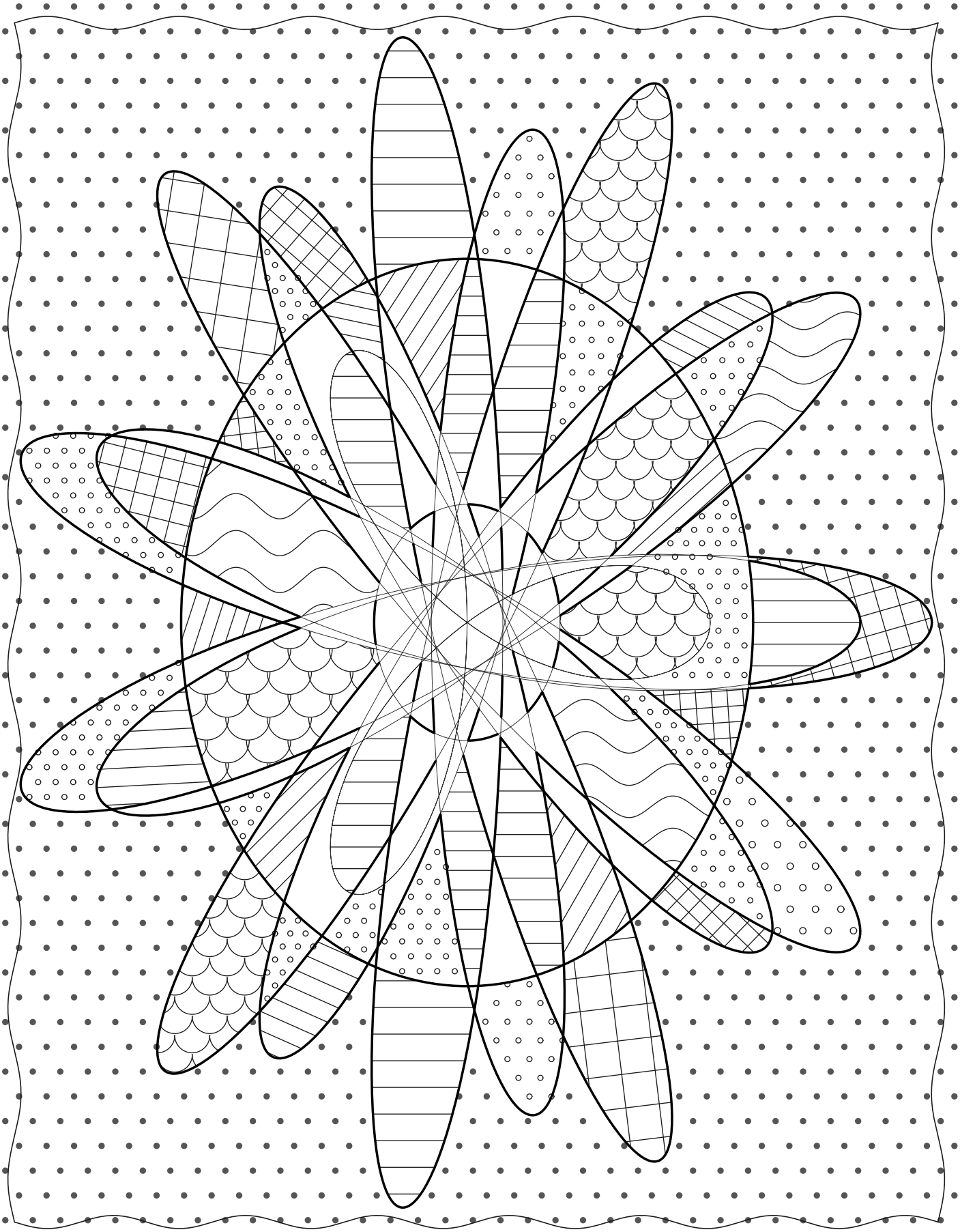


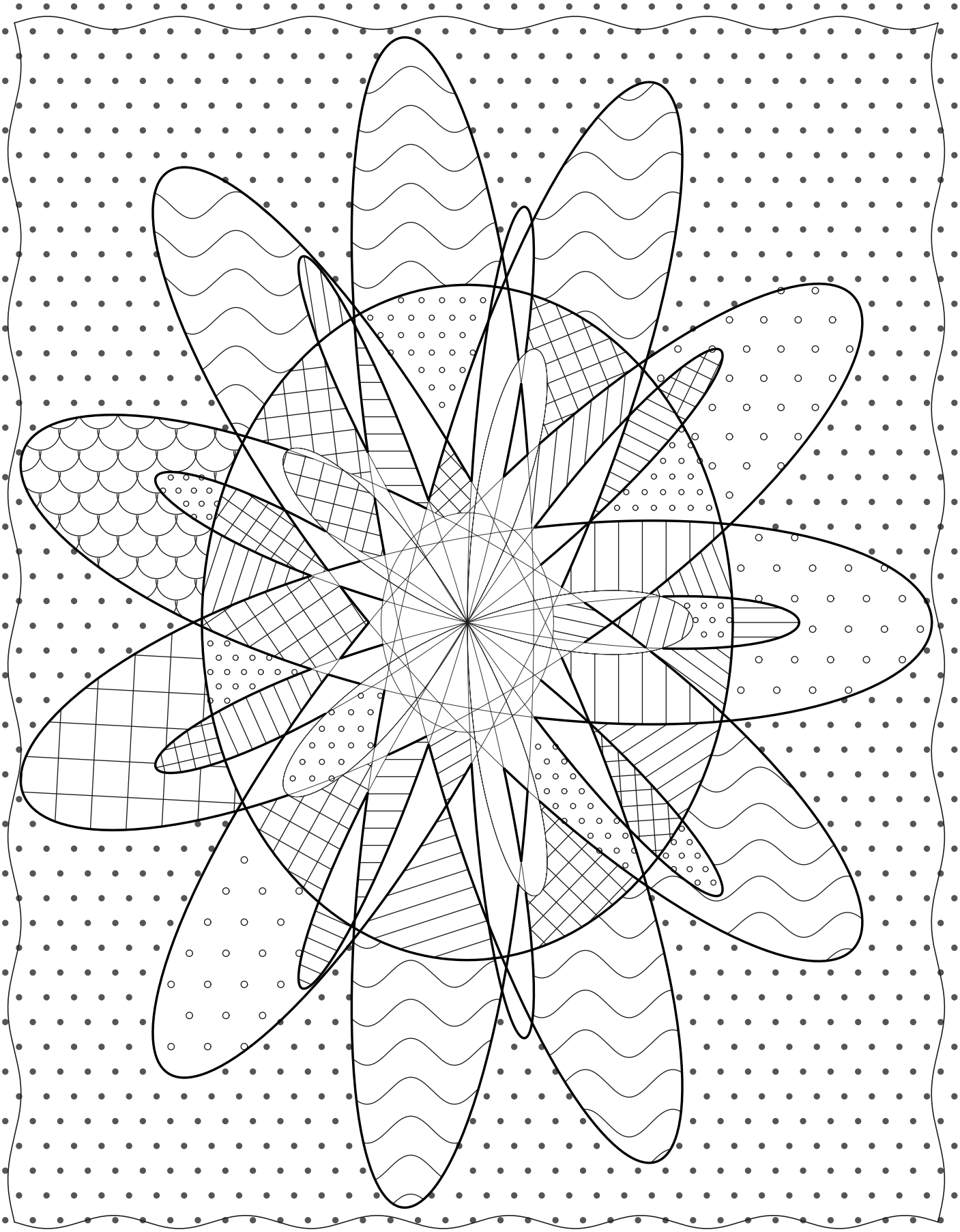


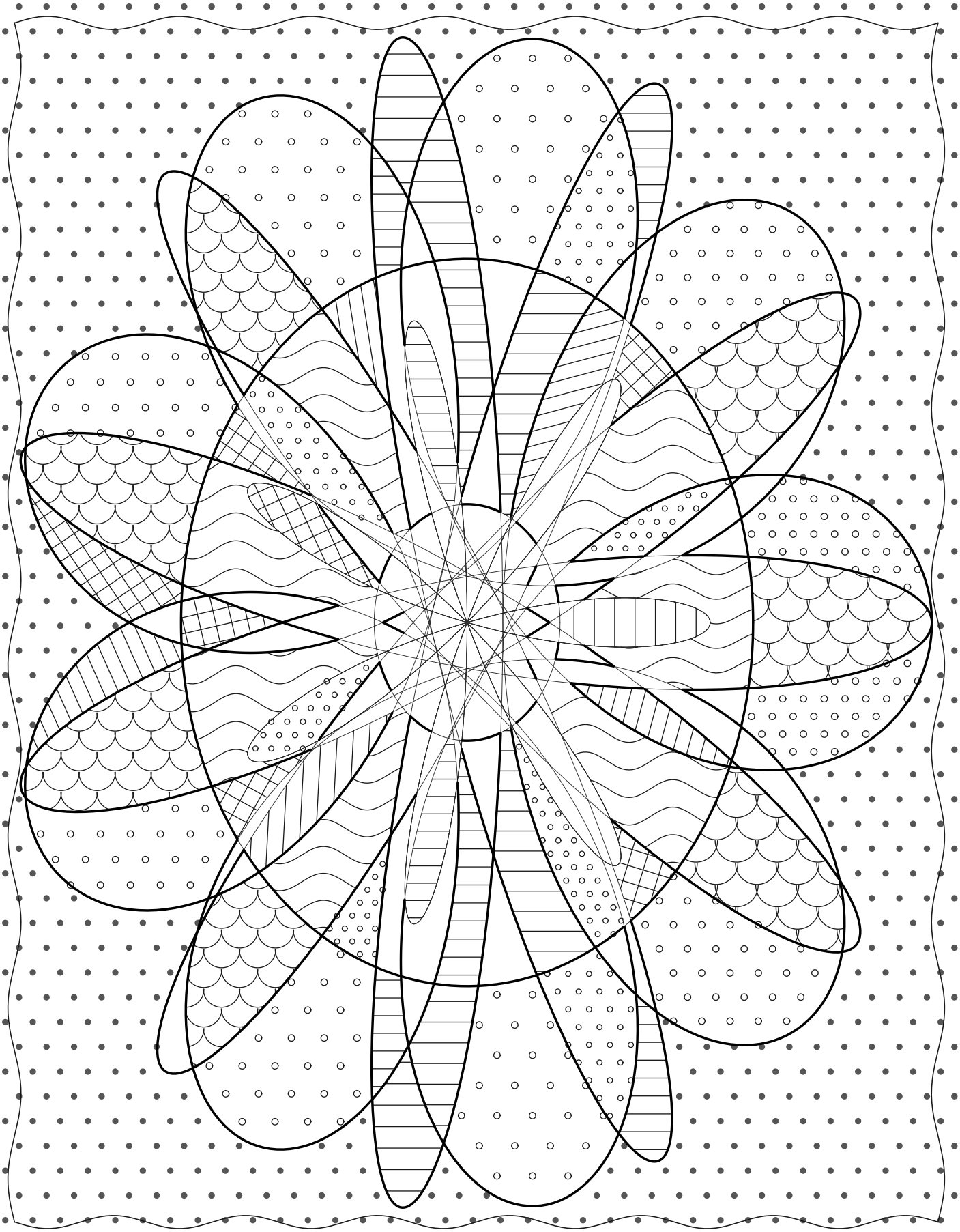


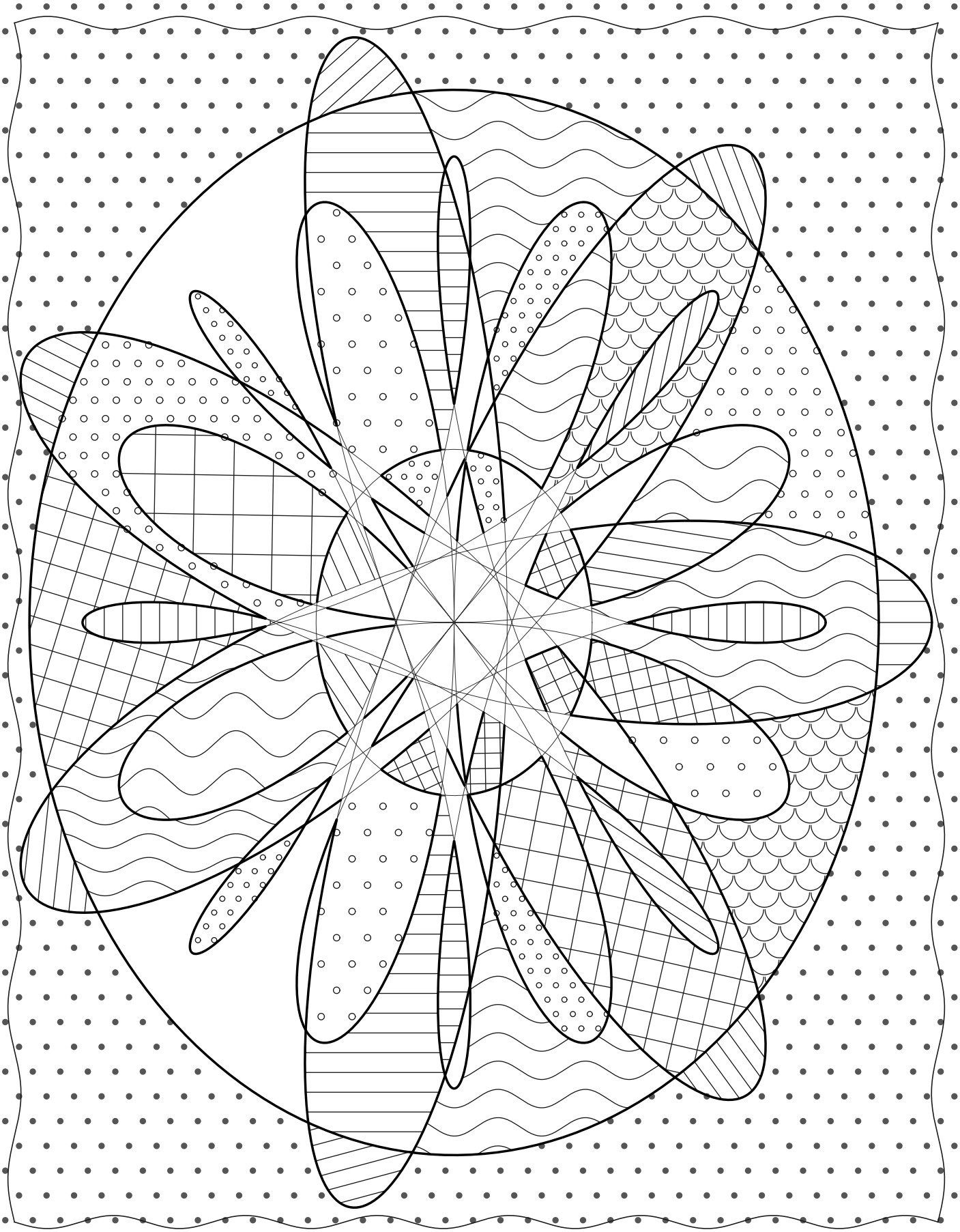


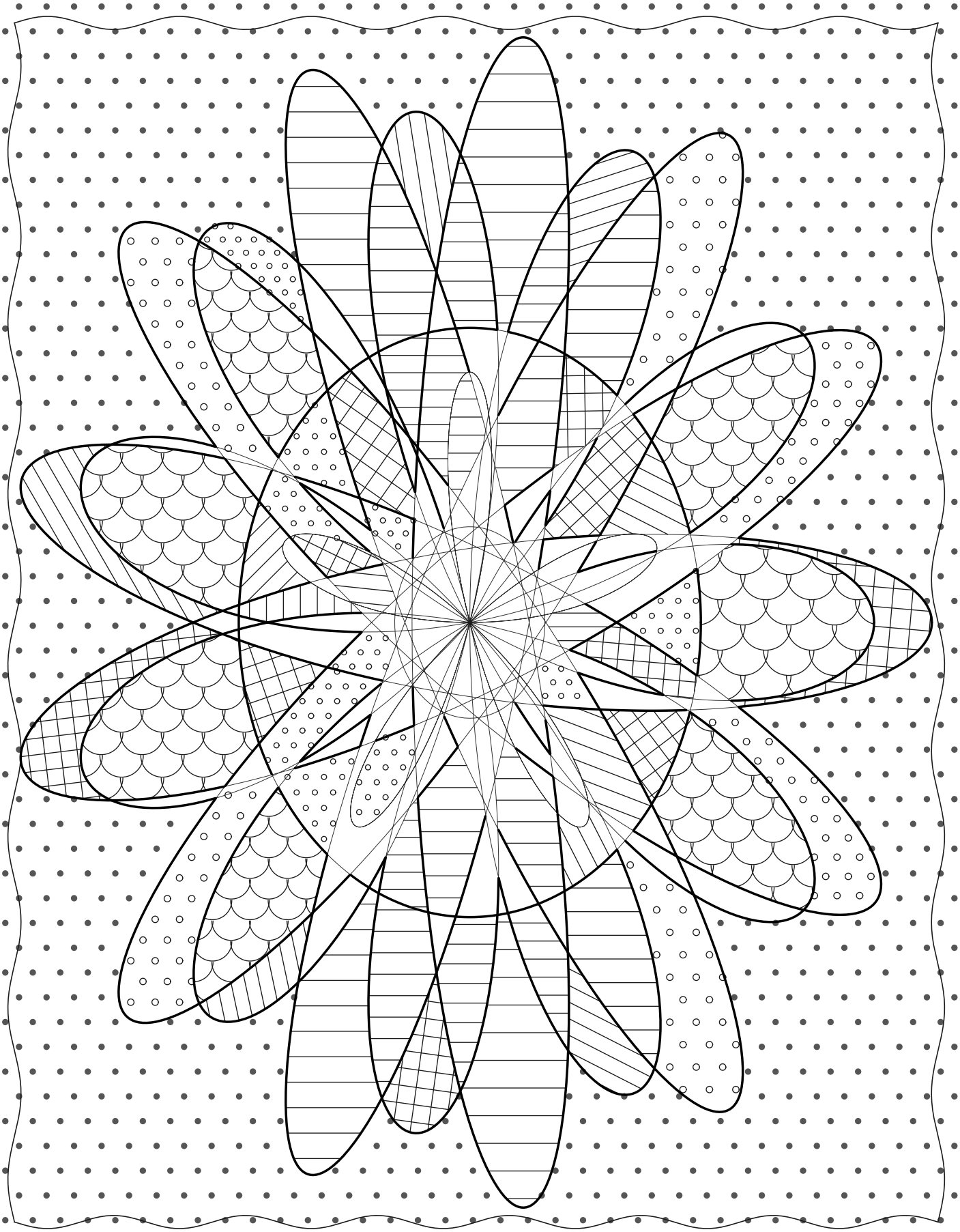


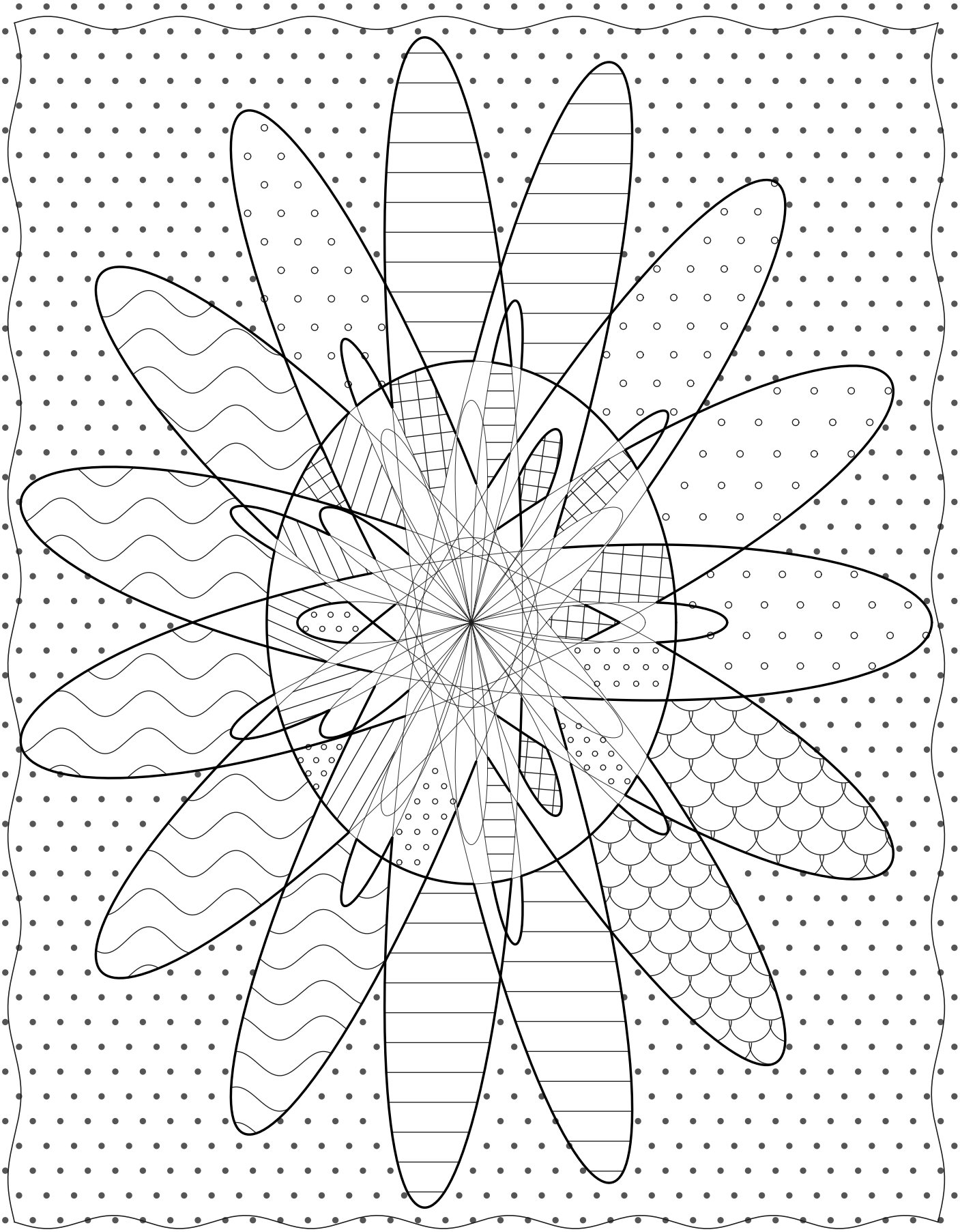


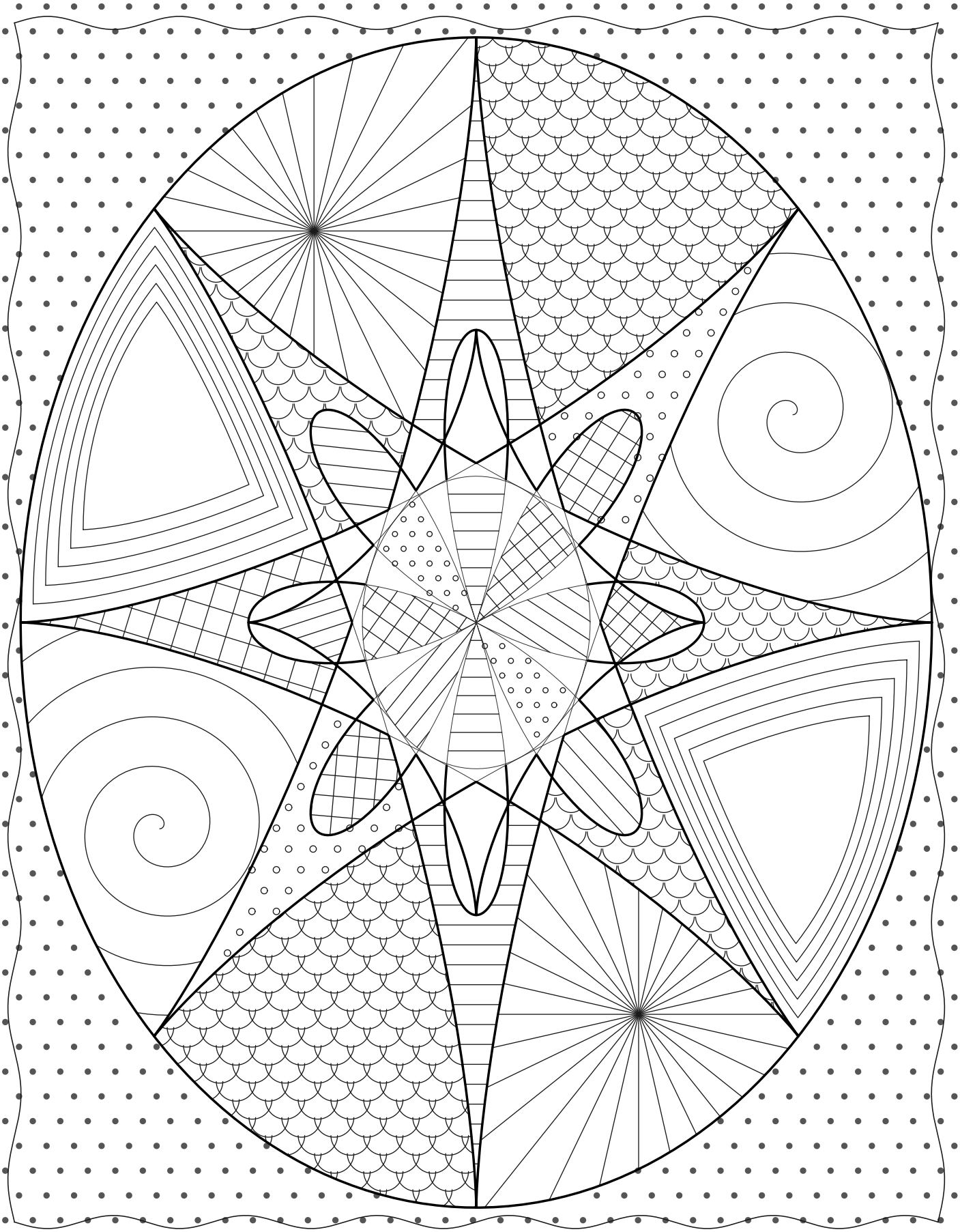


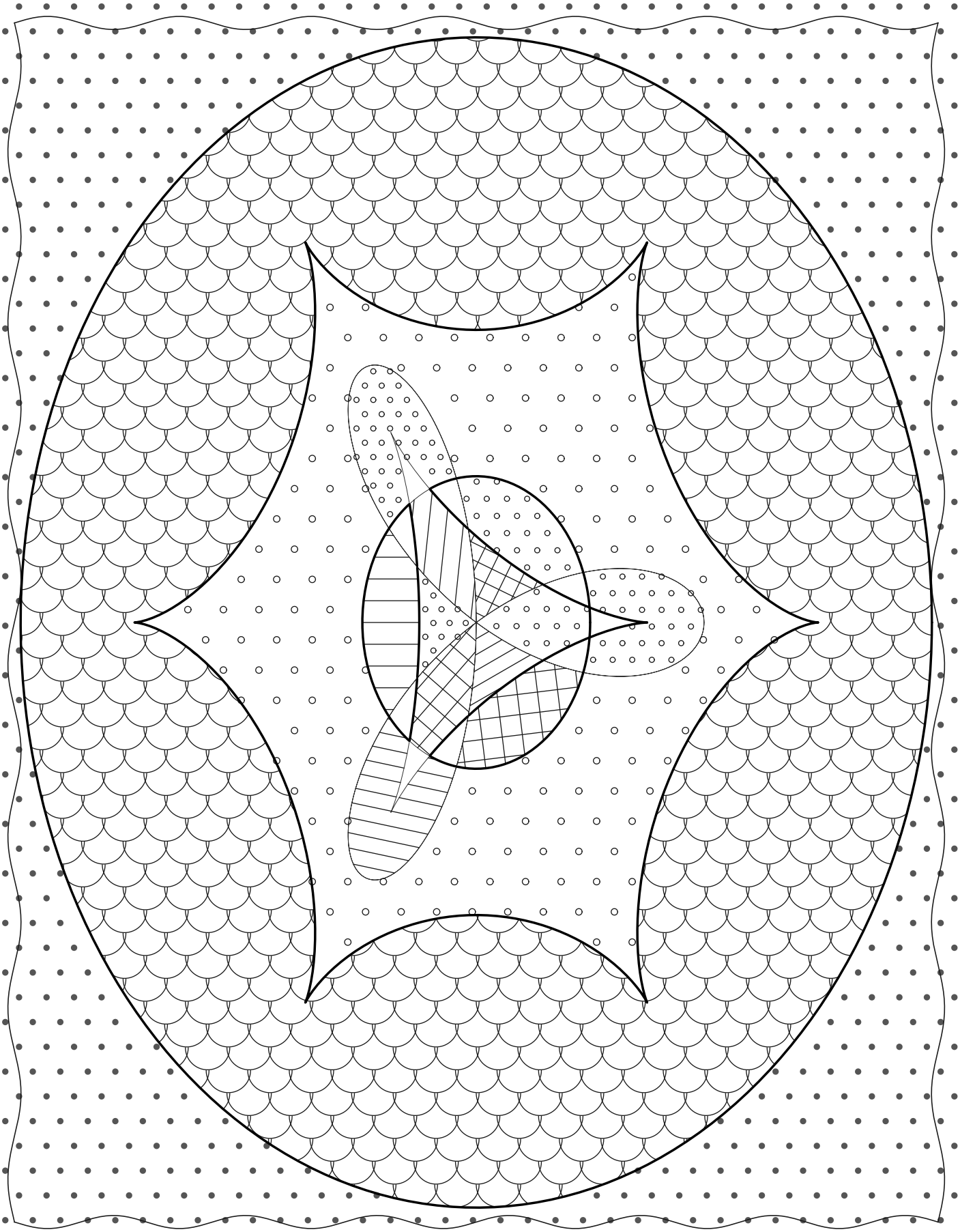


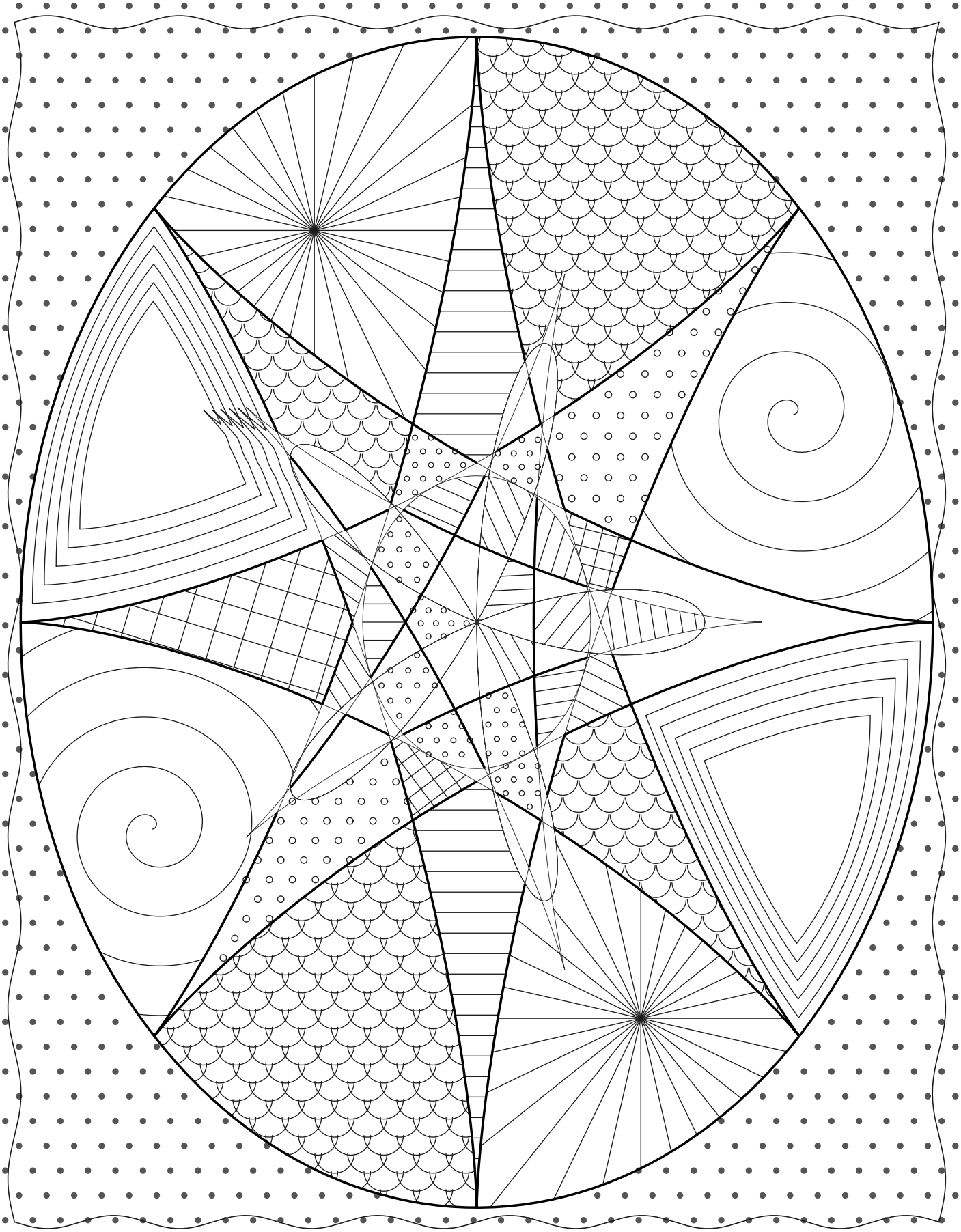


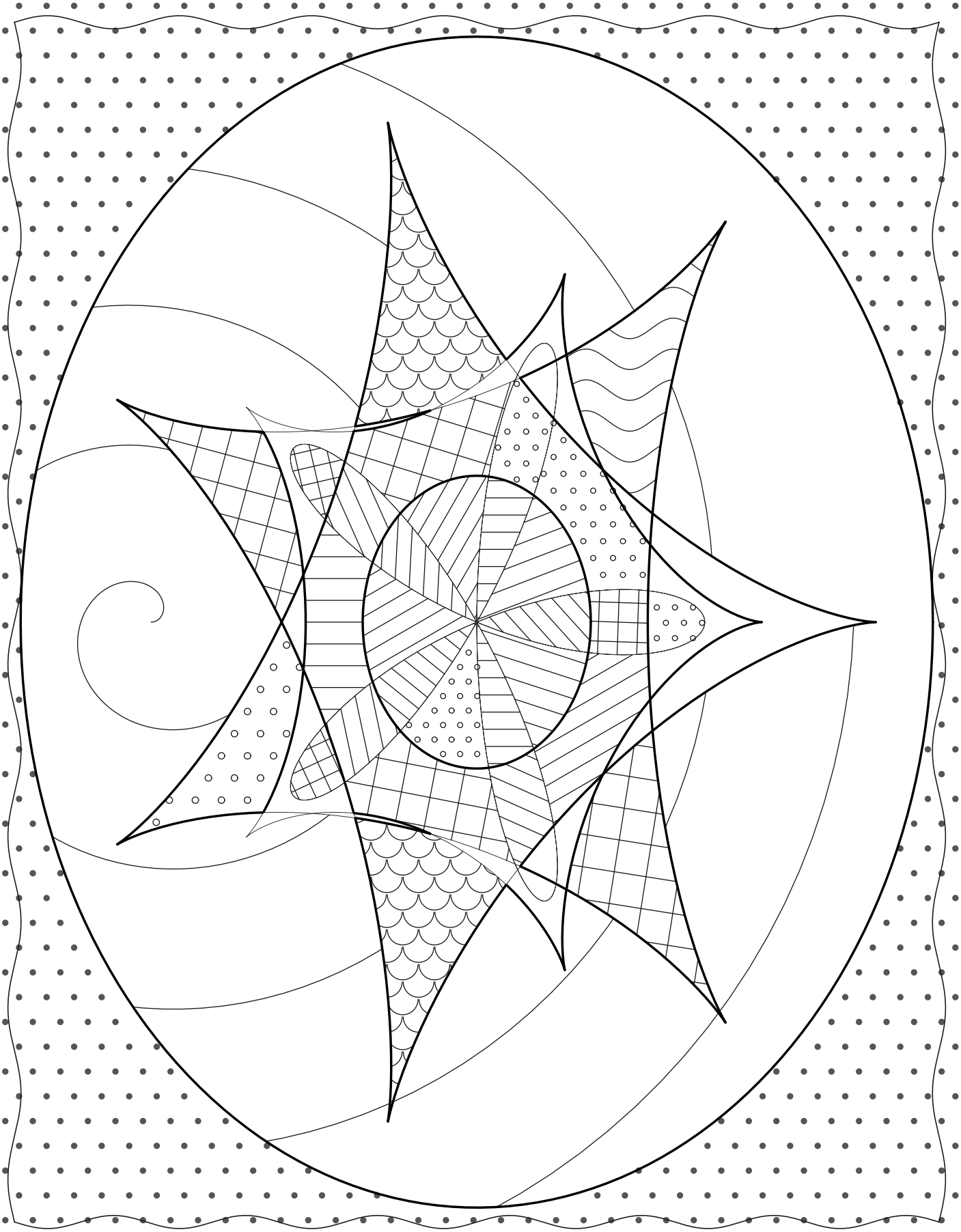


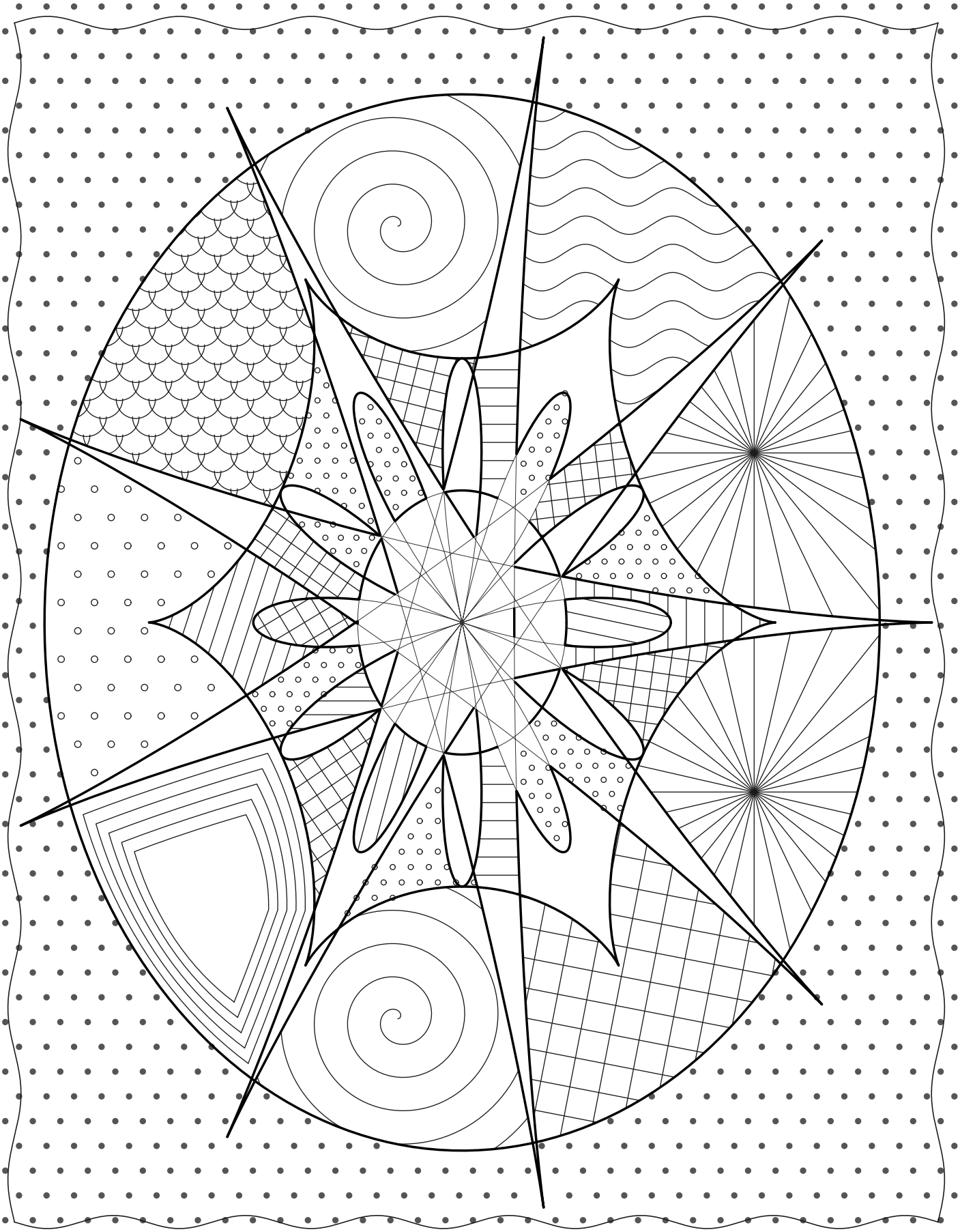


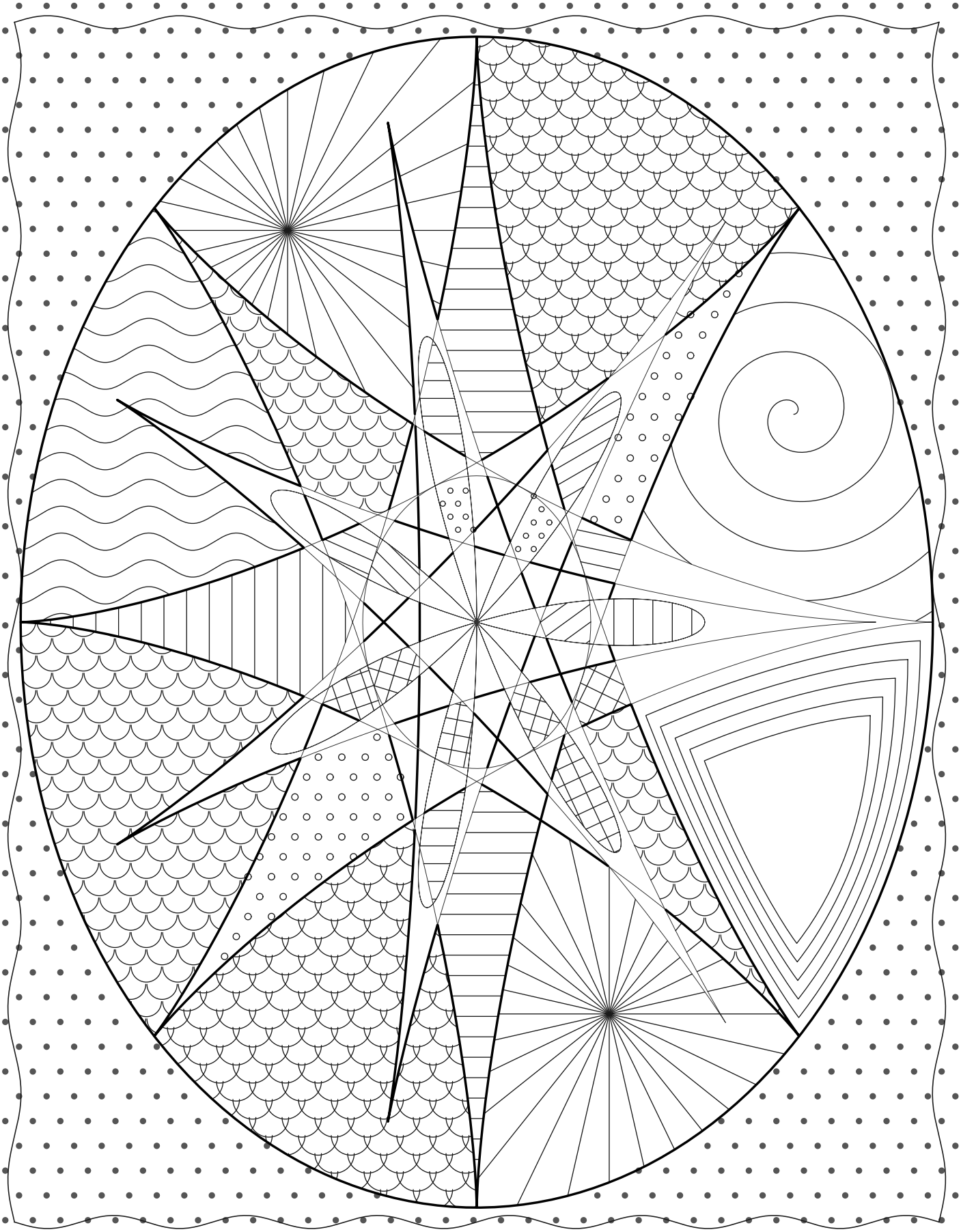


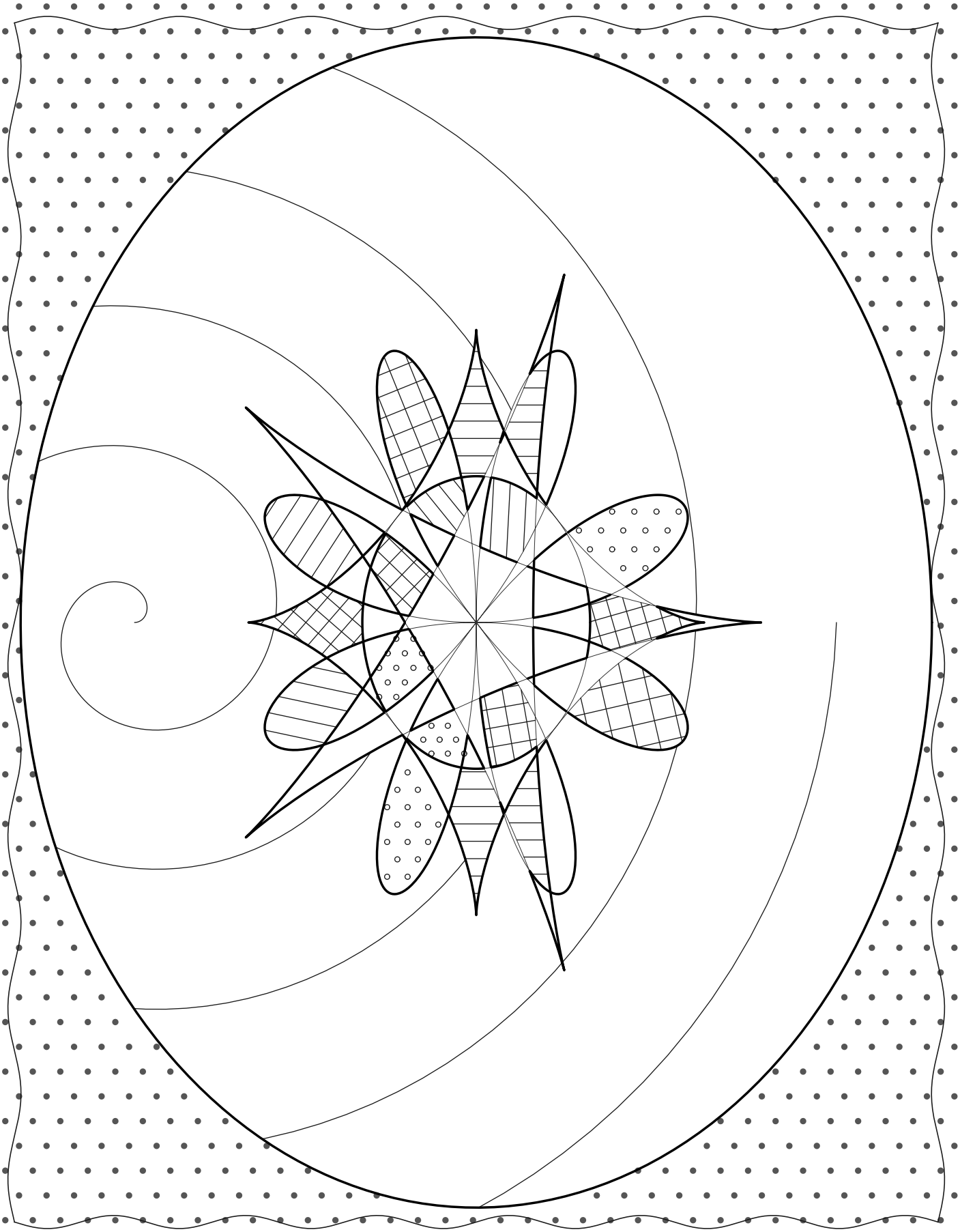


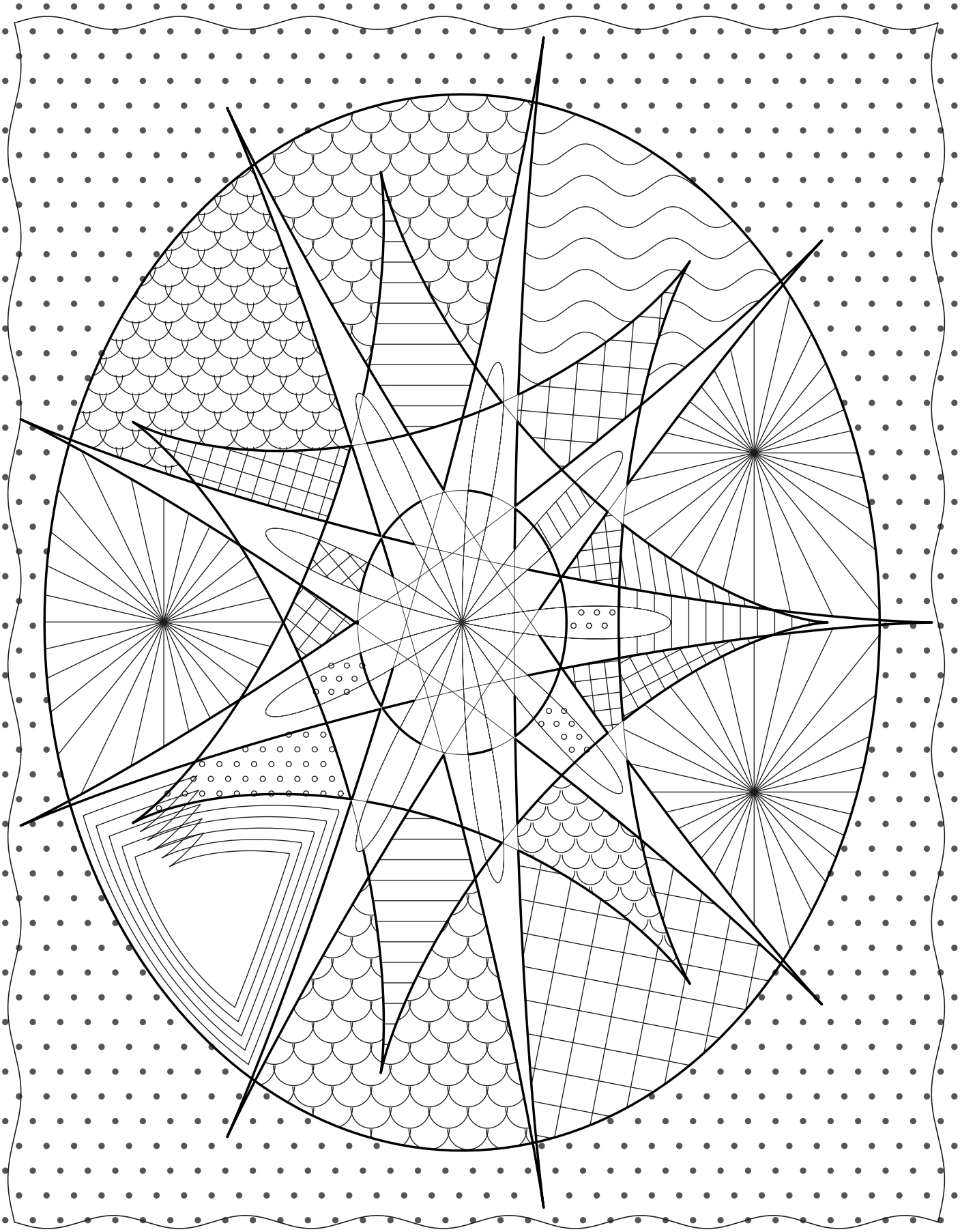


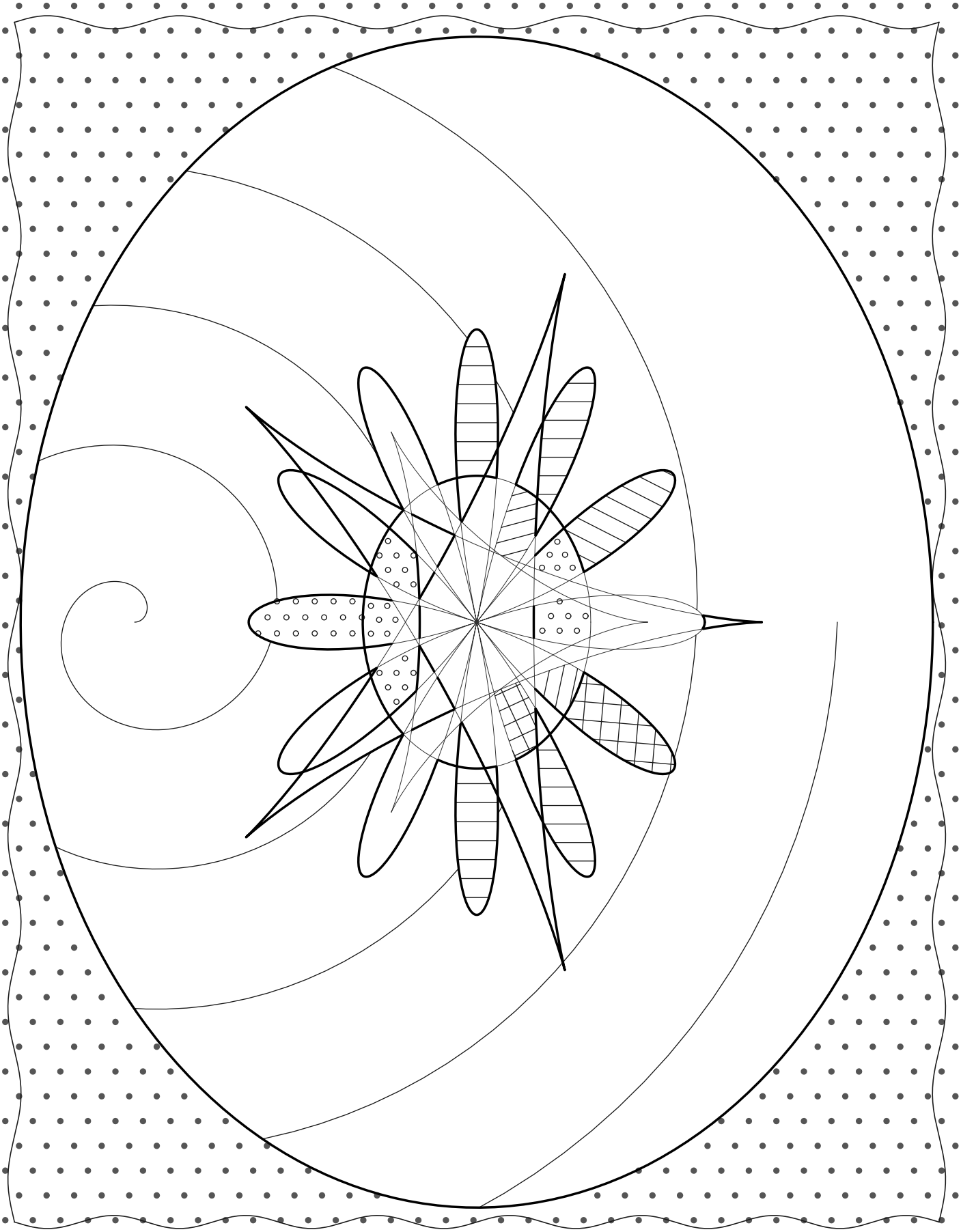


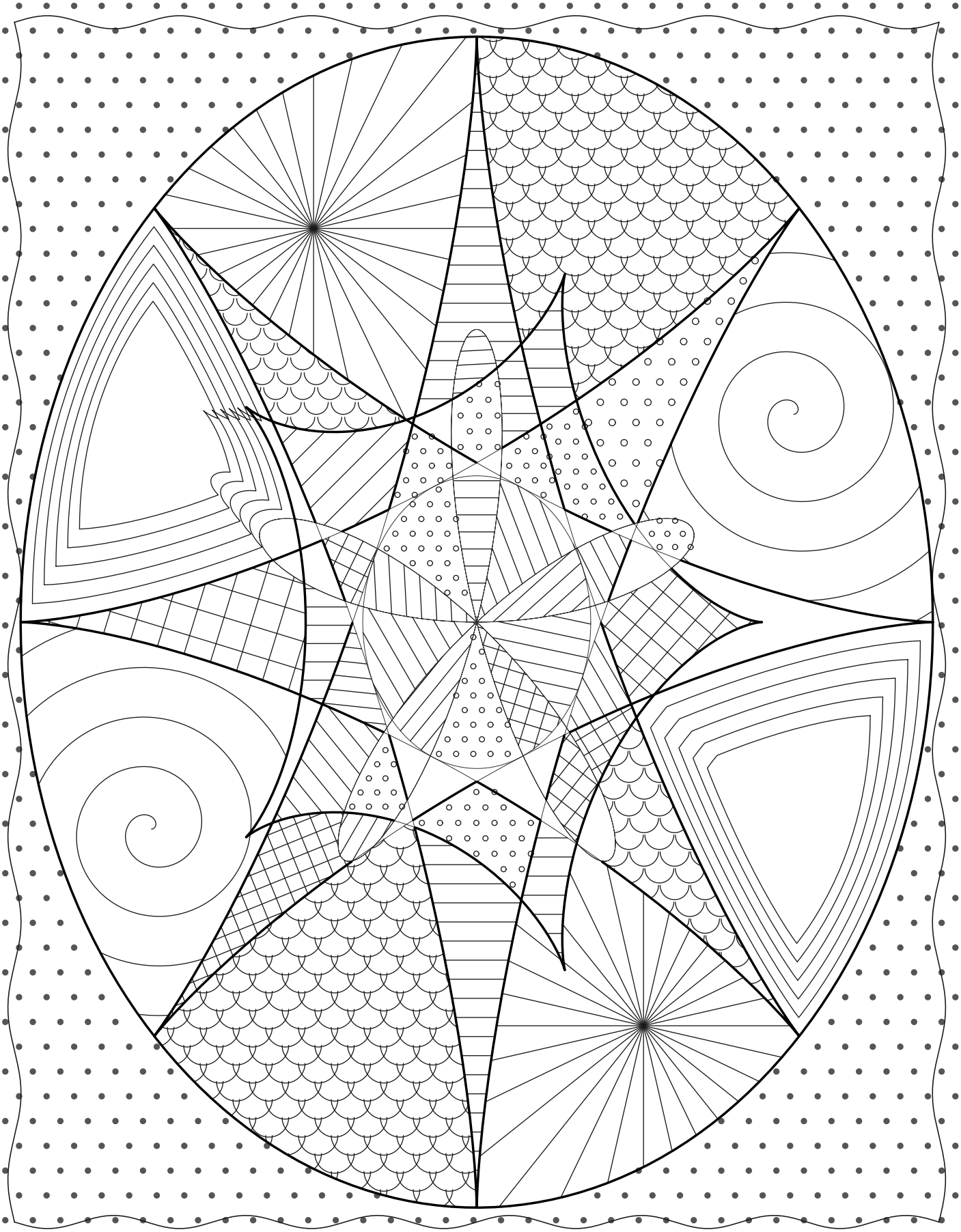


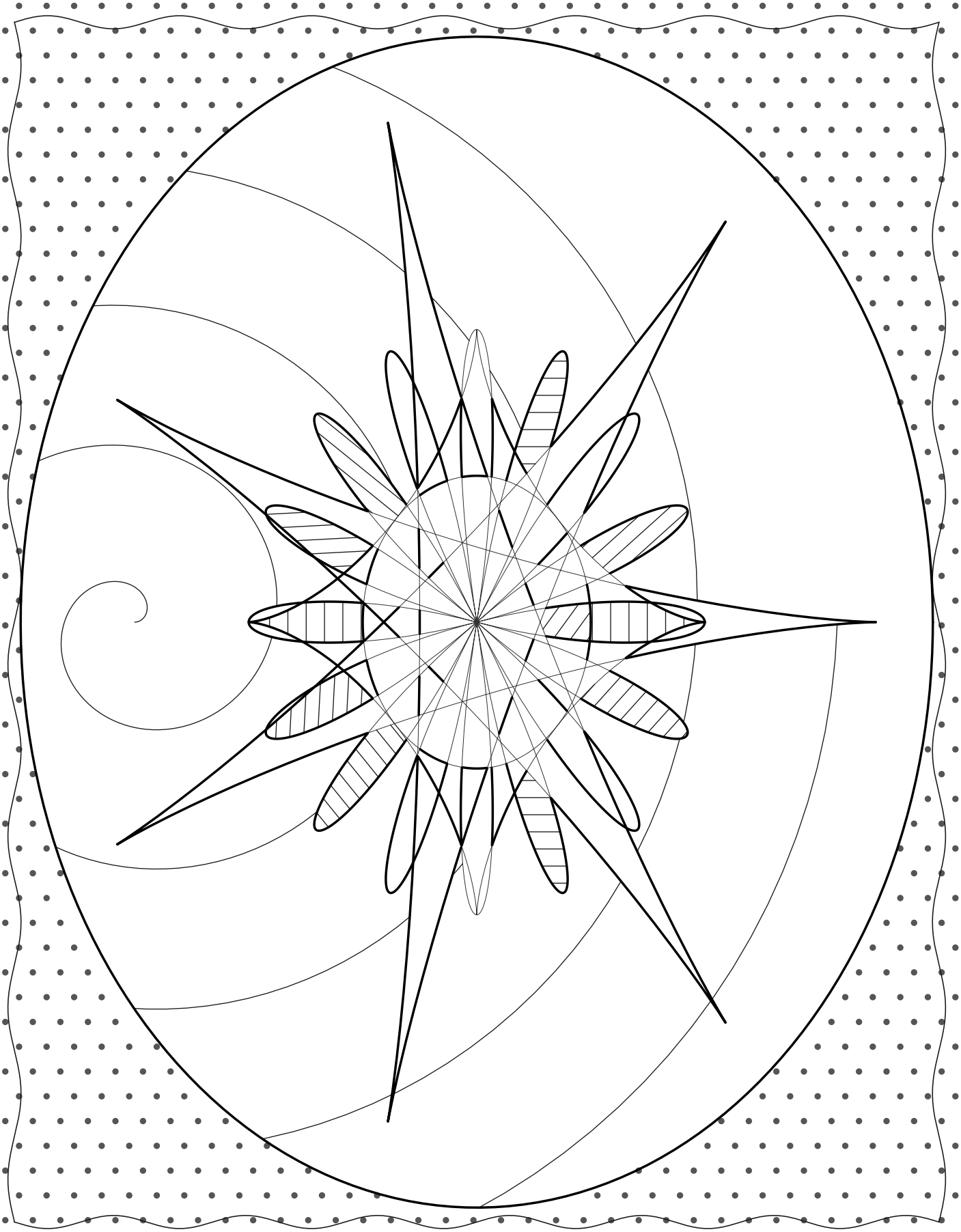


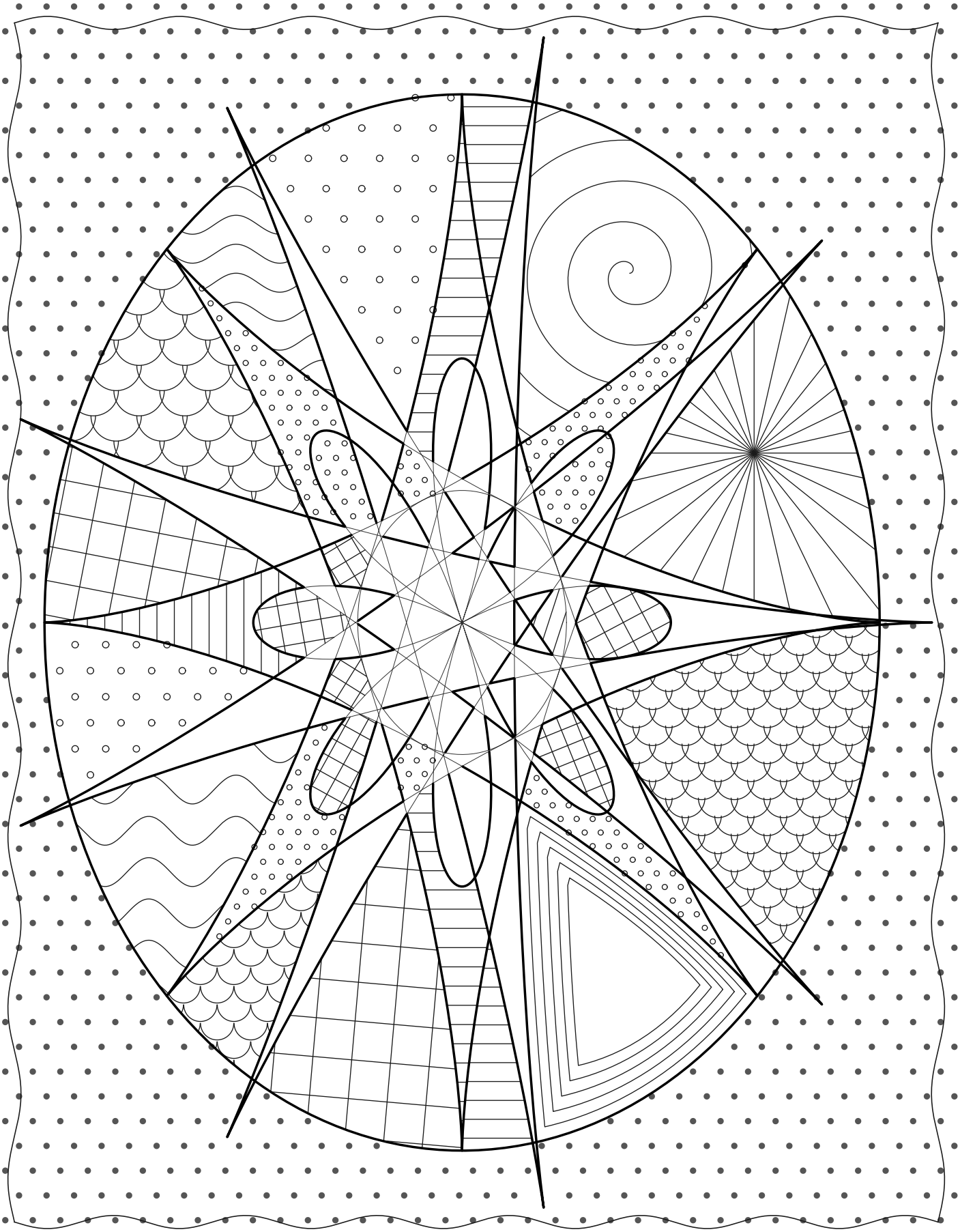


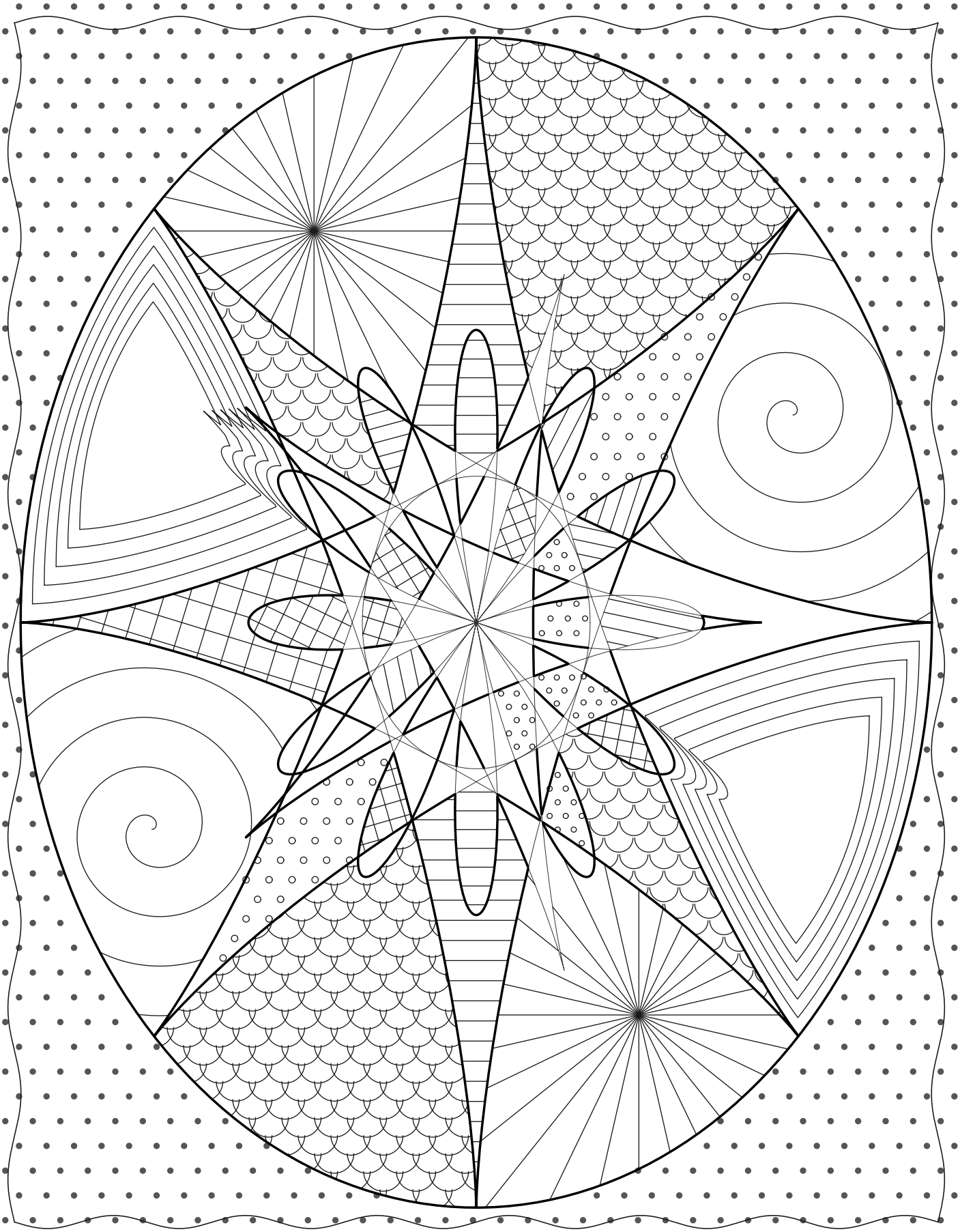


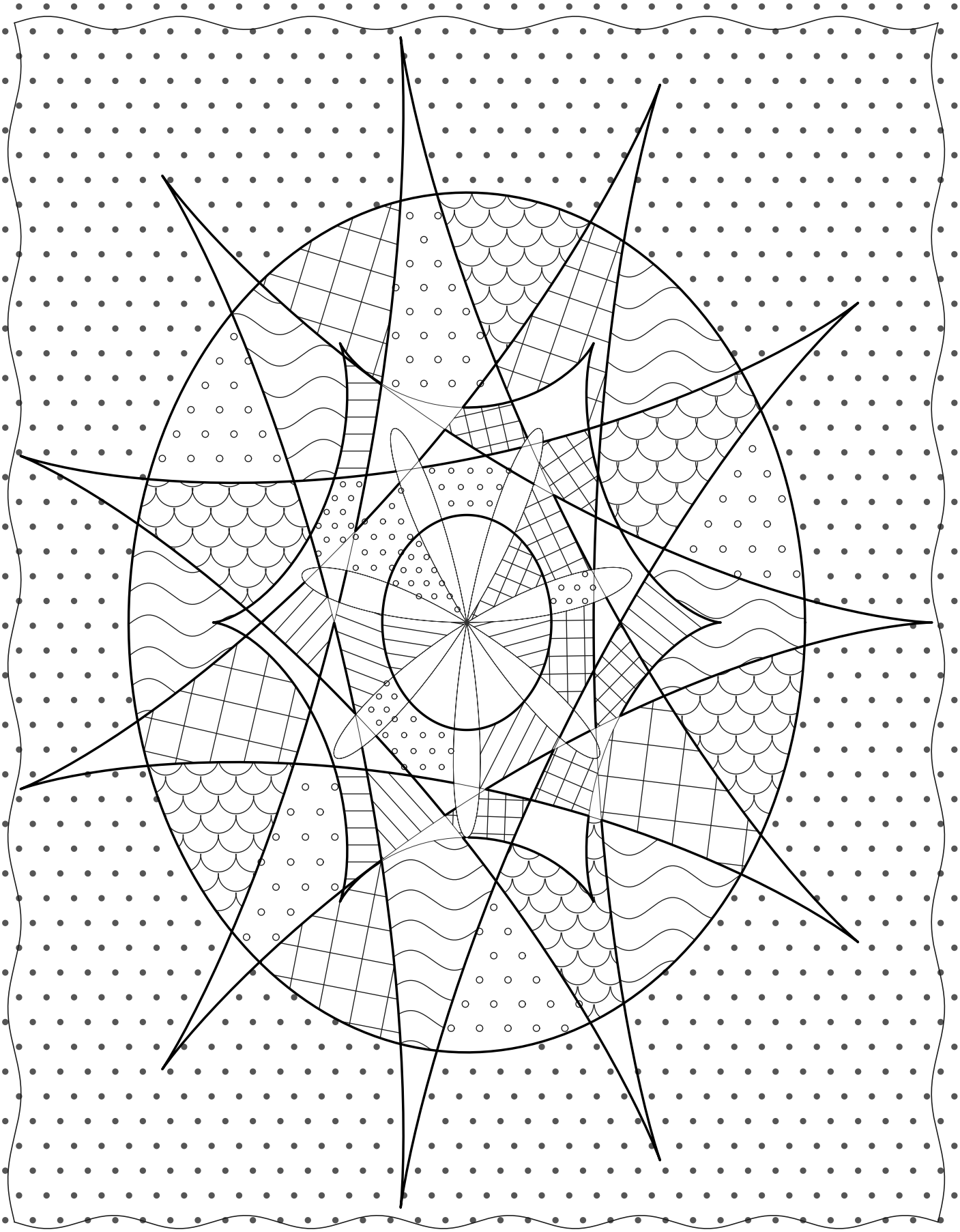


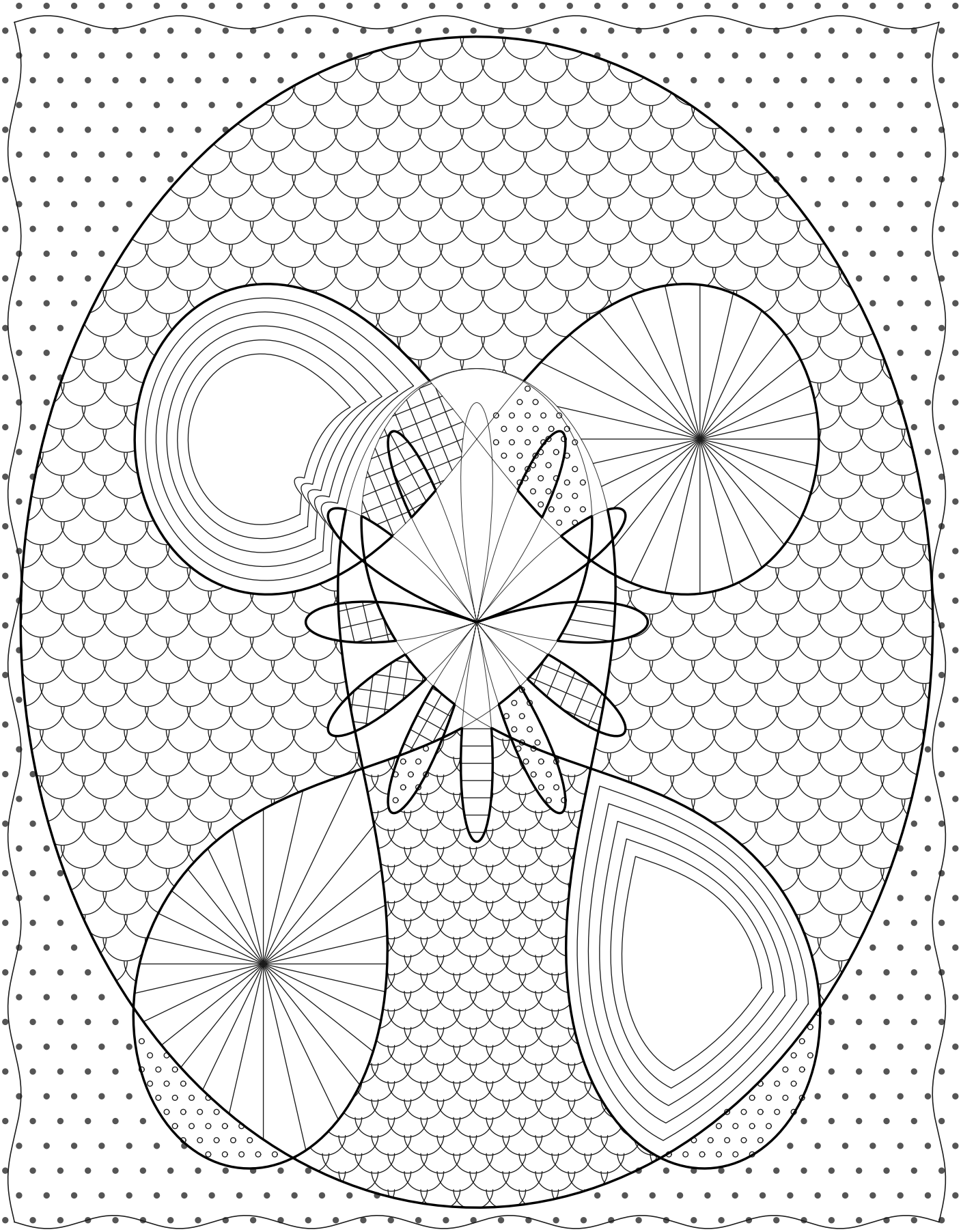


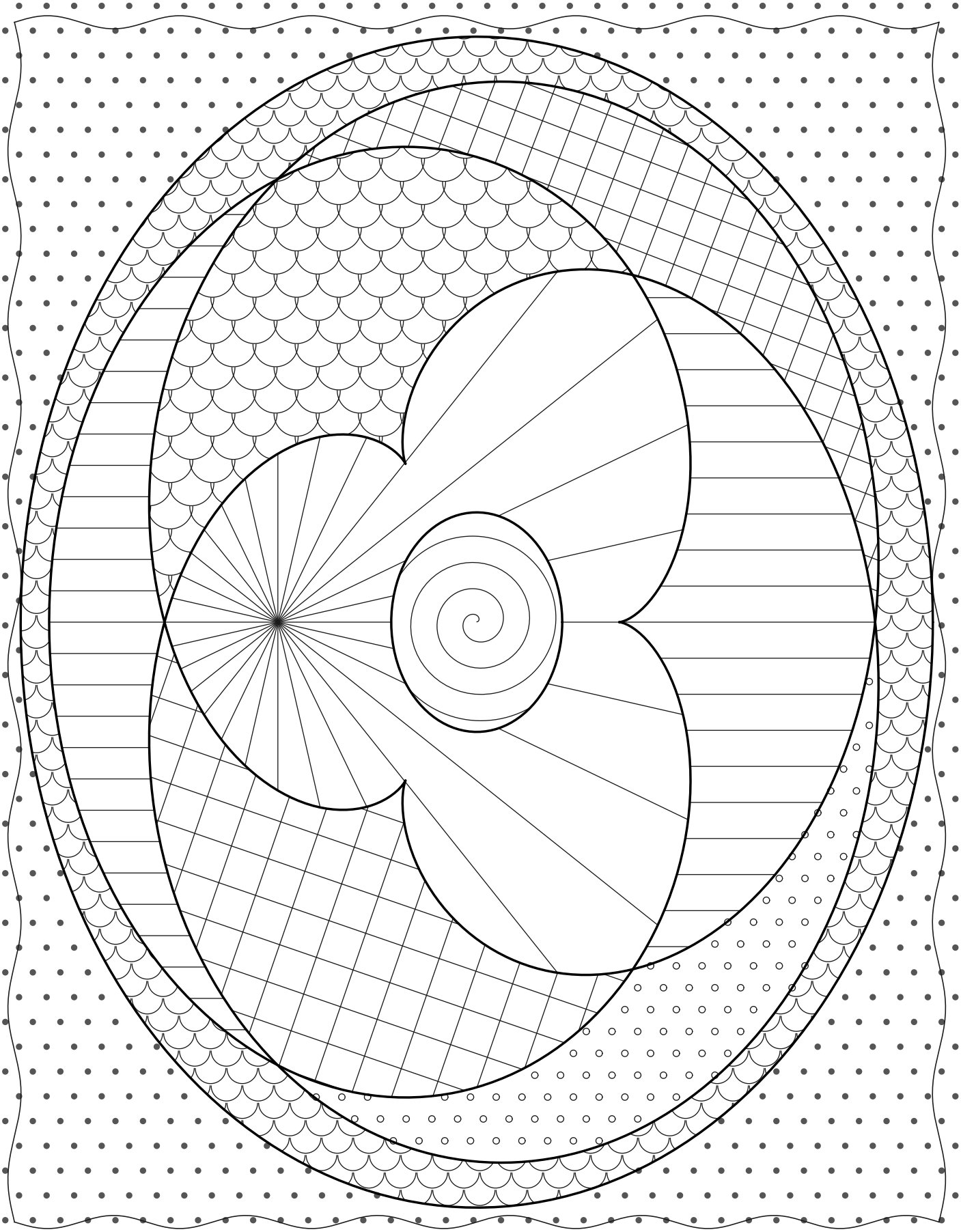


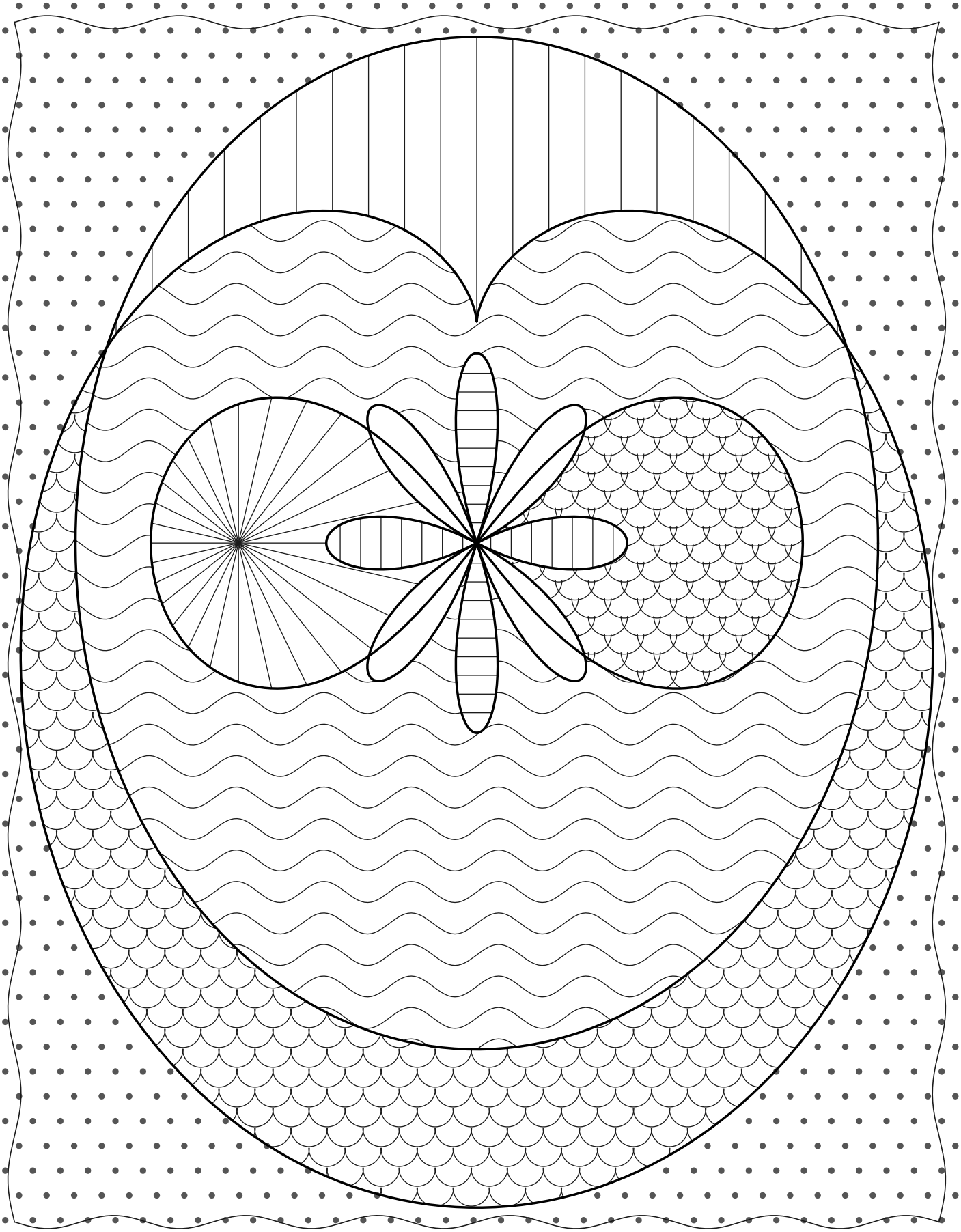


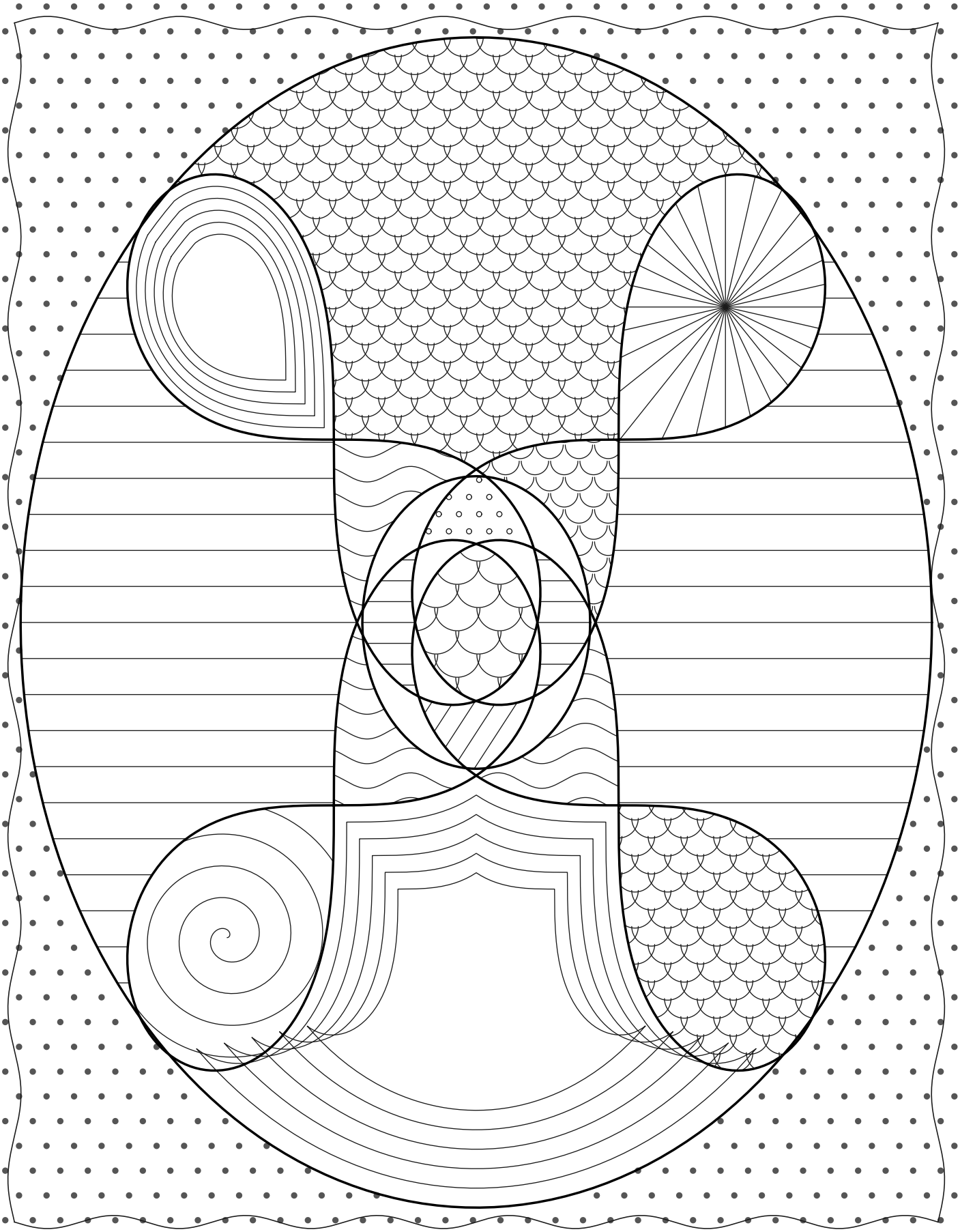


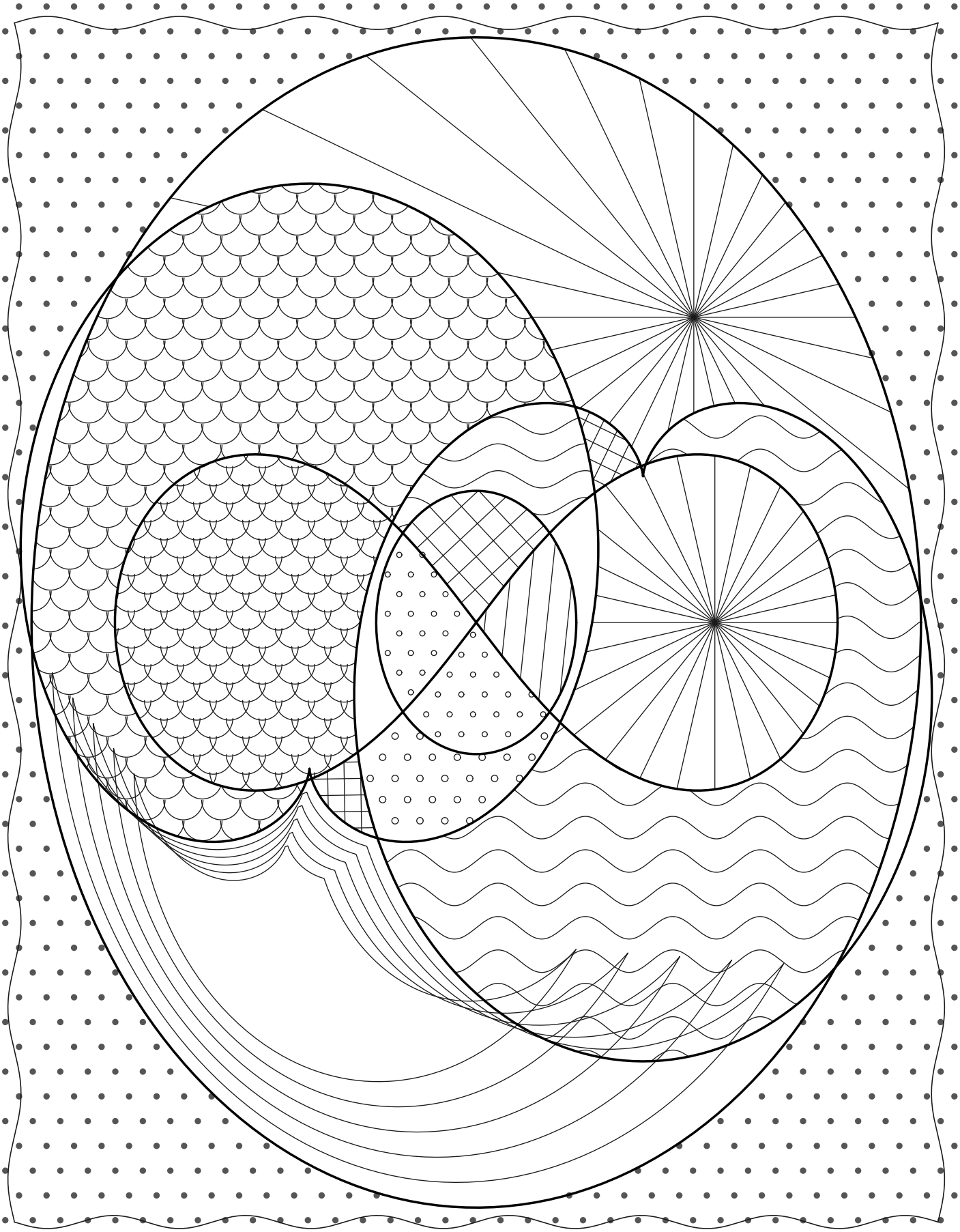


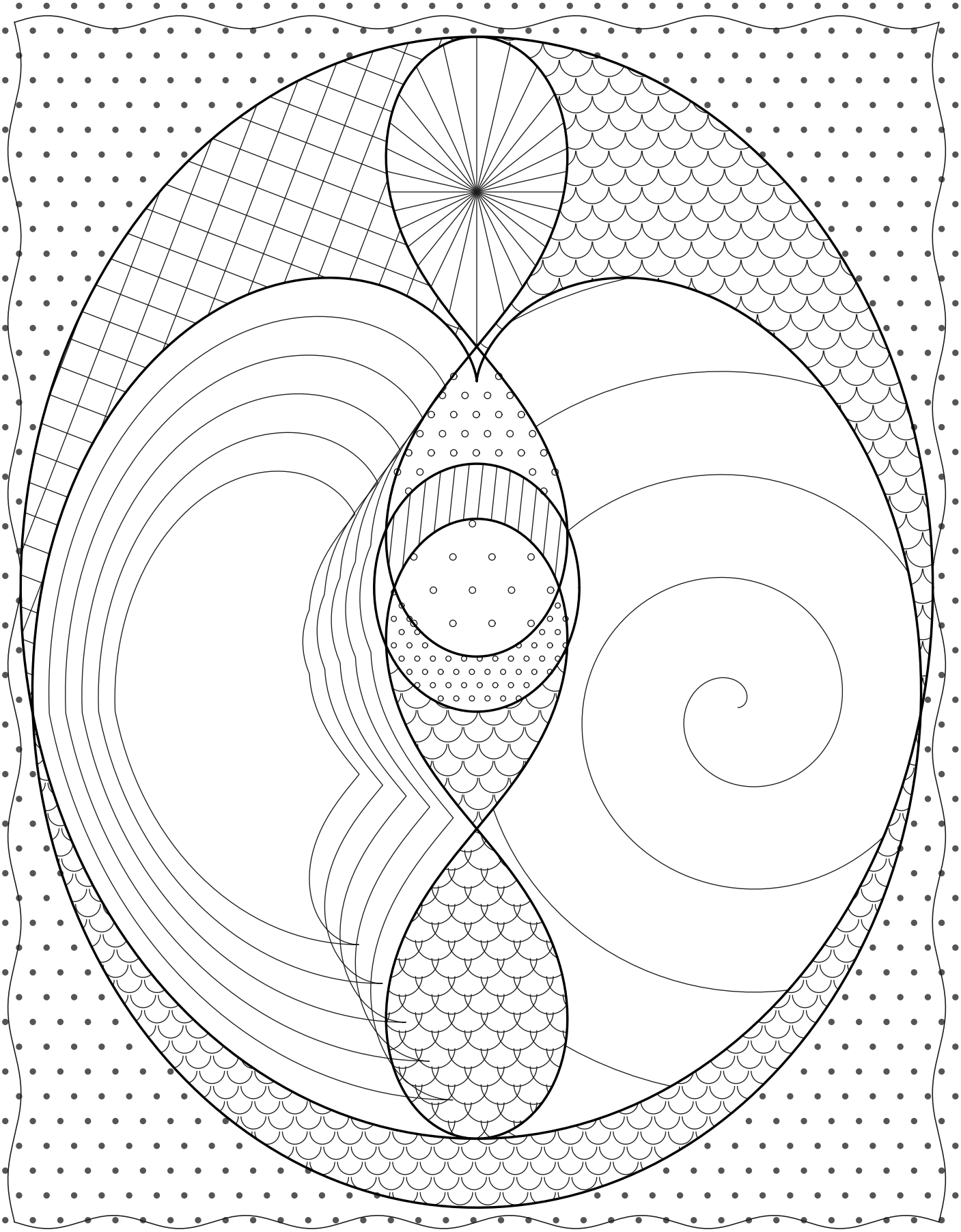


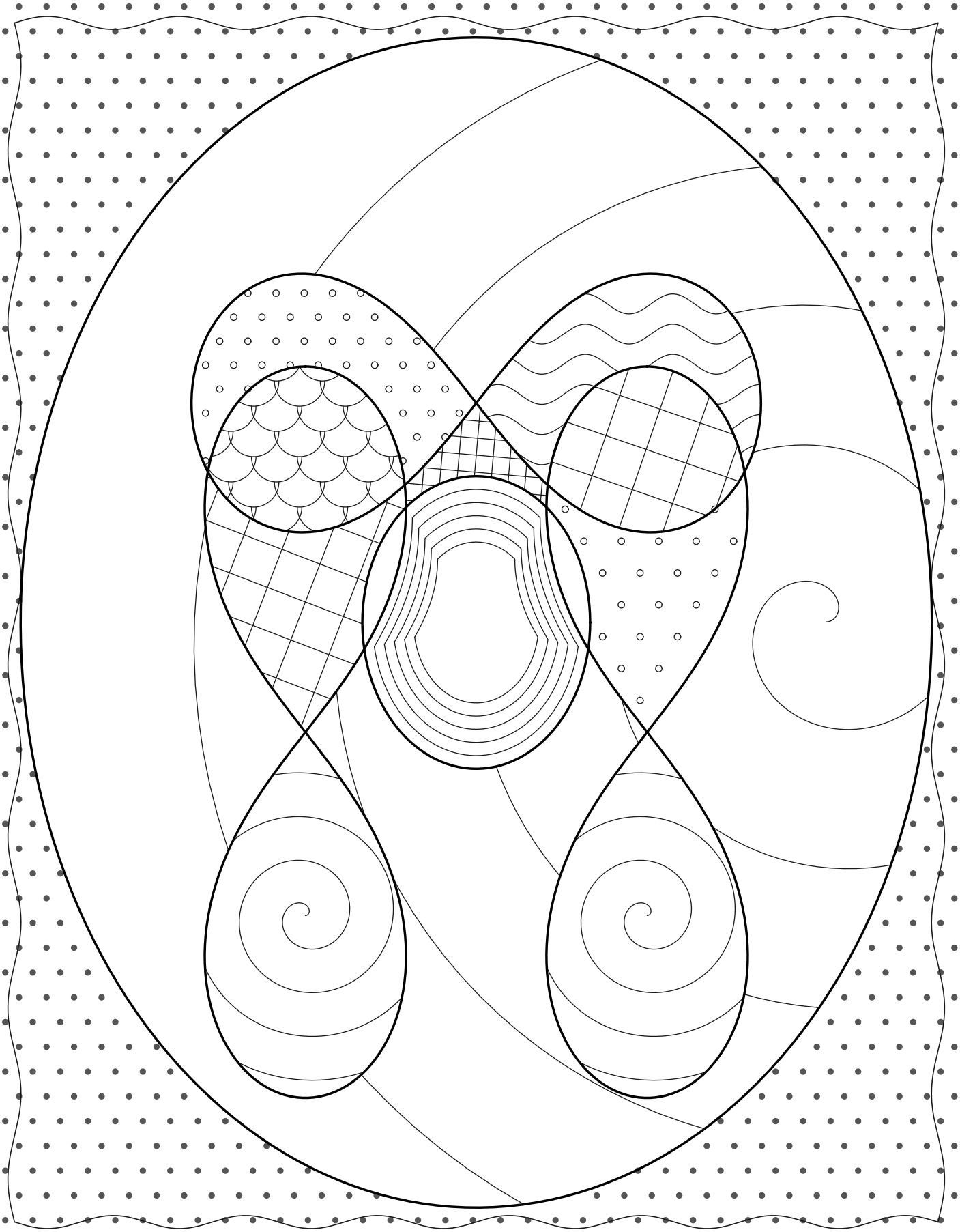


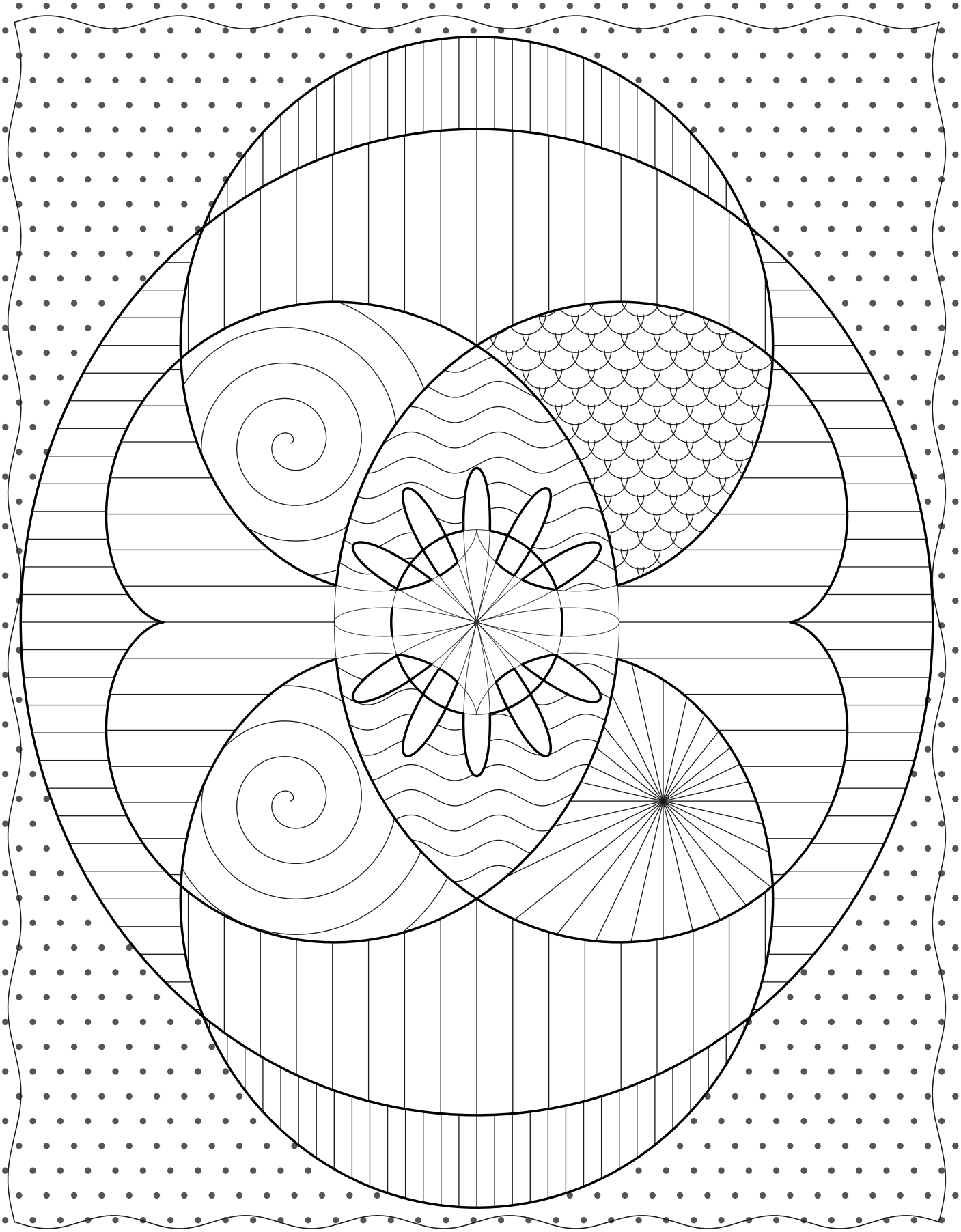






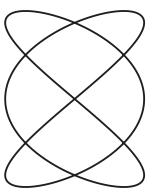






The Mathematics Behind the Art

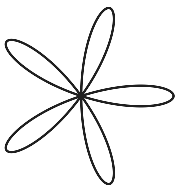
Every page in this book is the exact trace of a real mathematical curve. Here are the families you have been colouring.



Lissajous figure

$x = A \cdot \sin(a \cdot t + \delta) \qquad y = B \cdot \sin(b \cdot t)$

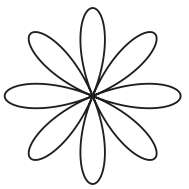
Two perpendicular vibrations. The frequency ratio a : b decides how the ribbon weaves and knots.



Polar rose

$r = a \cdot \cos(k \cdot \theta)$

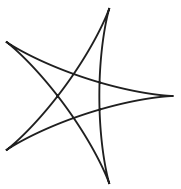
Petals open around the centre; the integer k sets how many bloom.



Spirograph (hypotrochoid)

$x = (R-r)\cos t + d \cdot \cos(((R-r)/r) \cdot t)$

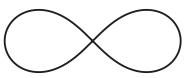
The path of a point fixed inside a small wheel rolling around the inside of a ring.



Hypocycloid

$x = (R-r)\cos t + r \cdot \cos(((R-r)/r) \cdot t)$

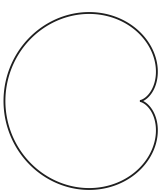
A wheel rolling inside a circle carves crisp, cusped stars.



Lemniscate of Bernoulli

$r^2 = a^2 \cdot \cos(2\theta)$

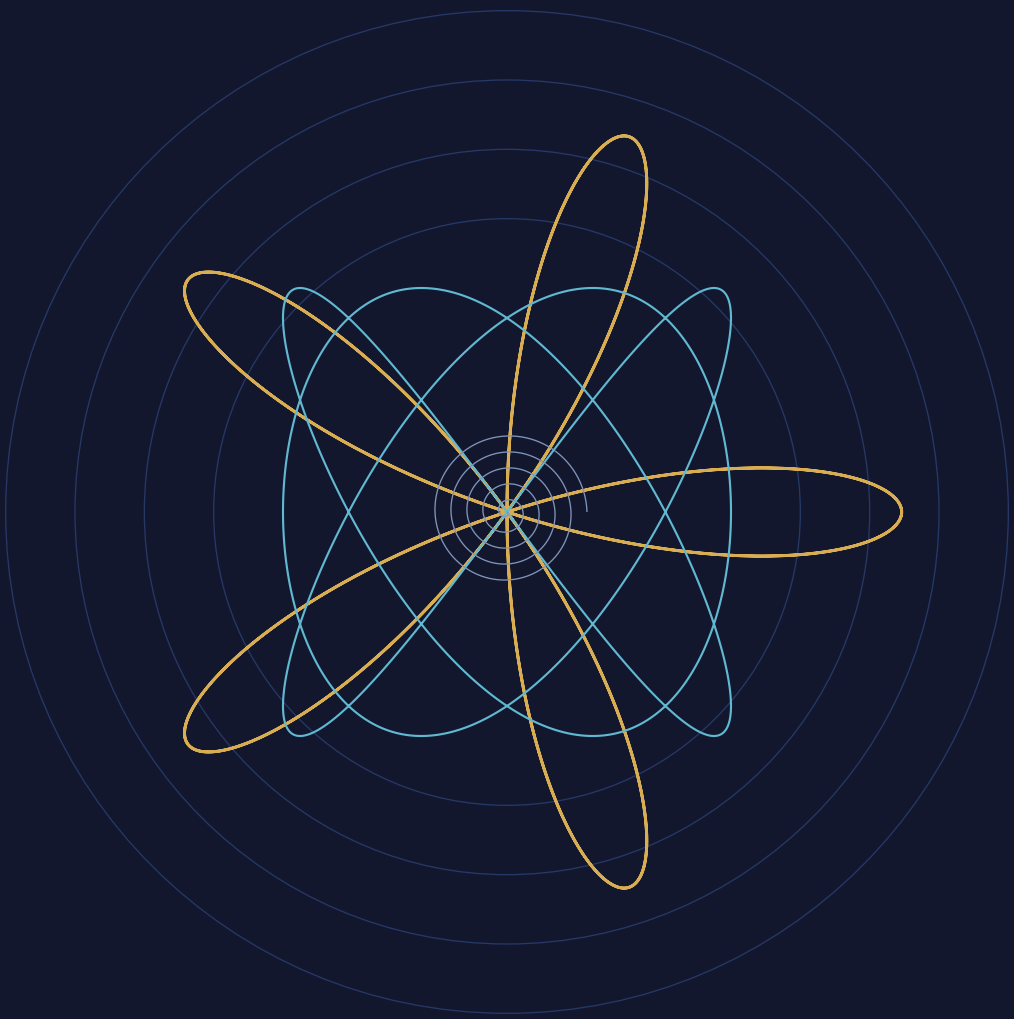
The elegant figure-eight — the mathematical infinity sign.



Cardioid

$r = a \cdot (1 - \cos\theta)$

A heart traced by one circle rolling around another the same size.



MathRelaxing

We hope you enjoyed every curve and equation.

Shop on Etsy: etsy.com/shop/themathrelaxing

YouTube · Pinterest · TikTok · Instagram @MathRelaxing

© MathRelaxing ? For personal use only. Not for redistribution or commercial use.